

BUILDING INTERPRETABLE CLIMATE EMULATORS FOR ECONOMICS*

Aryan Eftekhari, Doris Folini, Aleksandra Friedl, Felix Kübler, Simon Scheidegger and Olaf Schenk

We introduce a framework for developing efficient and interpretable climate emulators for economic models of climate change. The paper makes two main contributions. First, we propose a general framework for constructing carbon cycle emulators for macroeconomic models. The framework is implemented as a generalised linear multi-reservoir (box) model that conserves key physical quantities and can be customised for specific applications. We consider three versions of the carbon cycle emulator, which we evaluate within a simple representative agent economic model: (i) a three-box setting comparable to DICE-2016; (ii) a four-box extension; and (iii) a four-box version that explicitly captures land use change. While the three-box model reproduces benchmark results well, and the fourth reservoir adds little, incorporating the impact of land use change on the carbon storage capacity of the terrestrial biosphere substantially alters atmospheric carbon stocks, temperature trajectories and the optimal mitigation path. Second, we investigate pattern-scaling techniques that transform global mean temperature projections from climate emulators into spatially heterogeneous warming fields. We show how regional baseline climates, non-uniform warming and the associated uncertainties propagate into economic damages.

JEL codes: C61, E27, Q5, Q51, Q54, Q58

There is a rapidly developing literature on the macroeconomics of climate change (for recent reviews, see, e.g., Dietz, 2024; Fernández-Villaverde *et al.*, 2025) that uses so-called climate emulators (CEs) within macroeconomic models to explore the economic consequences of industrial carbon emissions. These emulators are reduced-form models that map emissions into atmospheric CO₂, global mean temperature and regional warming, while reproducing the salient behaviour of earth system models (ESMs; e.g., Geoffroy *et al.*, 2013a).

Folini *et al.* (2025) scrutinise the three-box climate emulator embedded in DICE-2016 (Nordhaus, 2017) and develop a calibration-validation protocol for that specific structure. Their contribution, however, stops short of asking whether the three-box architecture itself is adequate and how emulator uncertainty propagates when regional impacts are required. In this paper, we address these two questions that are pivotal for contemporary models in climate economics.

- (i) *Carbon cycle resolution.* Does replacing the DICE-2016 three-box representation with richer multi-reservoir models materially alter carbon stocks, temperatures and optimal policies?

* Corresponding author: Simon Scheidegger, Department of Economics, University of Lausanne, Quartier de Chambronne, Internef 509, 1015 Lausanne, Switzerland. E-mail: simon.scheidegger@unil.ch

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- (ii) *Spatial downscaling*. How robust is standard pattern-scaling when the economic model demands grid-level or regional climate inputs (see Desmet and Rossi-Hansberg, 2024)? Specifically, how much additional uncertainty does the downscaling step inject beyond the already considerable uncertainty in global CO₂ concentrations and temperatures (Santer *et al.*, 1990; Tebaldi and Arblaster, 2014; Lynch *et al.*, 2017)?

To answer these questions, we introduce an open-source toolbox that lets economists design physically consistent linear box model carbon cycle emulators (CCEs), then auto-calibrate and rigorously test them. The same framework links to pattern-scaling libraries, keeping emulator, pattern and baseline uncertainties in a transparent manner.

Integrated assessment models (IAMs) often employ a single linear rule that maps cumulative emissions to global temperature change (e.g., Dietz and Venmans, 2019, and references therein). This cumulative-emissions climate emulator is computationally attractive because it collapses climate dynamics into a single stock of cumulative CO₂, but this parsimony masks four critical weaknesses: temperature change in response to changing atmospheric CO₂ is not instantaneous, but takes about ten years (Stjern *et al.*, 2023); forcings from non-CO₂ greenhouse gases or aerosols are relevant, but are ignored (Jenkins *et al.*, 2021); the non-distinction between land and ocean carbon reservoirs prevents any study addressing their relative role in climate or economic terms; the assumed linear relation negates a priori the existence of any non-linearities as expected, for example, in the context of negative emissions or climate tipping points (Lenton *et al.*, 2008; Zickfeld *et al.*, 2021). Consequently, relying on this linear relationship can lead to misleading temperature projections and, as a direct consequence, flawed policy guidance.

A second strand of literature employs two separate modules to predict the link between emissions and temperature: the former maps emissions into atmospheric concentration by means of a CCE, and the latter maps concentrations to temperature. The present paper focuses on the link between emissions and concentration. For this, there exist two alternative modelling frameworks. The literature adopts either multi-reservoir box models (Nordhaus, 2017; Dorheim *et al.*, 2024) or impulse-response formulations (IRFs; e.g., Meinshausen *et al.*, 2011; Gasser *et al.*, 2017; Leach *et al.*, 2021). Comparison studies document the strengths and weaknesses of each family (Nicholls *et al.*, 2021; Melnikova *et al.*, 2023). The two approaches can be shown to be formally equivalent when modelling the decay of a single pulse of carbon emitted to the atmosphere (Li *et al.*, 2009; Raupach *et al.*, 2011; Raupach, 2013). Both approaches have to be calibrated against properly chosen benchmark data.

In this article, we focus on a box model CCE, where the global system is partitioned into a handful of well-mixed ‘boxes’ (reservoirs) between which carbon diffuses.¹ In a box model, the atmosphere, the ocean layers, the land biosphere and, occasionally, additional reservoirs such as permafrost or the shallow ocean are treated as homogeneous pools connected by fluxes that obey mass balance, as illustrated in a stylised fashion in the ‘climate’ bubble of Figure 1 for one concrete, illustrative example. Along with the number of reservoirs, the number of free parameters and characteristic timescales increases. We argue that the box model formulation of the CCE is intuitive in its interpretation and adaptation to use-case-specific modifications, including accommodation of non-linearities (Hooss *et al.*, 2001; Glotter *et al.*, 2014; Zhang *et al.*, 2021). The box approach naturally captures feedbacks that arise because carbon emissions

¹ In most IAMs, a two-reservoir energy-balance module handles the evolution of the temperature of the atmosphere and the ocean. Since this aspect of climate emulators is extensively discussed in Folini *et al.* (2025), we do not address it in this paper.

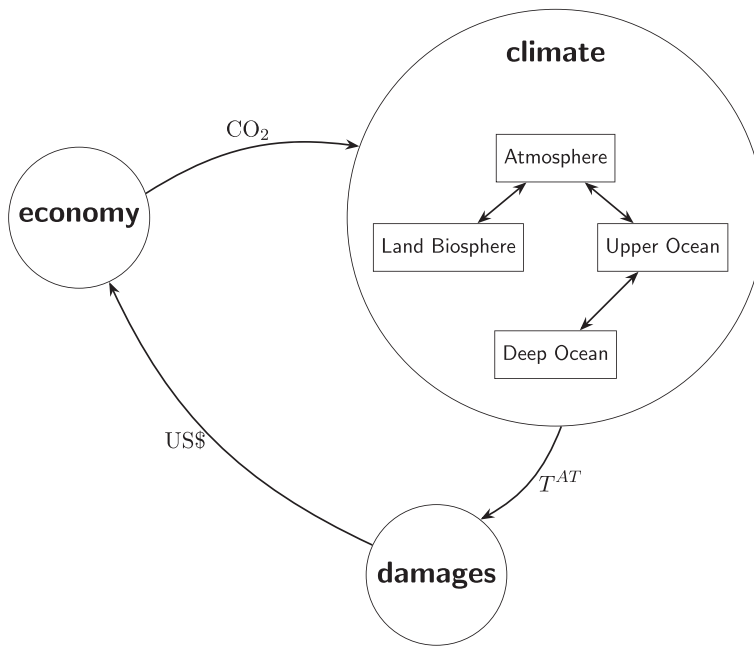


Fig. 1. *Stylised Schematic of an IAM.* In their structural form, an IAM in macroeconomics typically links three components: (i) an economic module that generates fossil fuel CO_2 emissions; (ii) a CE, that is, a reduced-form physical model that maps those emissions into atmospheric CO_2 concentrations and global mean temperature T^{AT} ; and (iii) a damage module that transforms the resulting temperature trajectory into economic losses (measured, e.g., in US\$). The inset within the CE visualises one particular CCE: a four-reservoir box model that tracks CO_2 exchanges among the atmosphere, land biosphere, upper ocean, and deep ocean. Double-headed arrows indicate the bidirectional carbon fluxes between these reservoirs. For clarity, the separate climate sub-component that links radiative forcing to temperature is not depicted.

to the atmosphere perturb the equilibrium among the different carbon reservoirs, yet remains computationally light.

The canonical DICE-2016 emulator (Nordhaus, 2017; Folini *et al.*, 2025), for instance, considers a three-reservoir carbon cycle box model. The model contains no explicit land biosphere reservoir, even though terrestrial carbon stocks and biological carbon dioxide removal (CDR) dominate many low-cost mitigation pathways. Our central question in this paper is how modelling choices, such as how many carbon reservoirs to include, which datasets to fit, and which physical constraints to enforce, shape projected concentrations (and ultimately economic damages).

We present an open-source toolbox that lets economists design, calibrate, and deploy bespoke CCEs that are physically sound and policy-specific. A key advantage of this modular approach is that it makes modelling choices, how many reservoirs to include, which datasets to fit and which physical constraints to impose, both transparent and testable.

We construct three emulators, 3SR, 4PR and 4PR-X, of increasing complexity, tracing how each additional layer sharpens economic insight:²

² Acronyms indicate reservoir count and routing of carbon: 3SR has three serial reservoirs (atmosphere *arrow* upper ocean *arrow* deep ocean); 4PR adds a land biosphere reservoir in parallel with the upper ocean, but keeps its storage capacity fixed; 4PR-X builds on 4PR by allowing the land biosphere capacity to vary with land use, thus capturing afforestation and deforestation feedbacks.

- 3SR: a re-implementation of the DICE-2016 three-reservoir carbon cycle, used as our validation anchor;
- 4PR: a variant that adds a land biosphere box with *static* carbon-holding capacity; and
- 4PR-X: a new model in which the land reservoir holding capacity adjusts endogenously to land use change, capturing feedbacks from deforestation or afforestation.³

Three- and four-box emulators (3SR and 4PR), calibrated to both pre-industrial (PI) and present-day (PD) conditions, replicate the historical and long-run evolution of atmospheric CO₂ and global temperature to within about 5% and 3%, respectively, over a 500-year horizon, an accuracy widely regarded as ‘fit for purpose’ in policy analysis. Including a static land box, therefore, leaves the headline results essentially unchanged. To illustrate the extent to which the choice of a CCE, the associated calibration targets and the selected hyperparameters can influence quantitative results, we evaluate the three carbon cycle models within the standard DICE economy–climate framework under PI conditions and a range of extreme scenarios. While 3SR and 4PR produce broadly similar policy-relevant trajectories, the 4PR-X variant, which permits the land biosphere to evolve dynamically under deforestation or urbanisation, yields markedly higher atmospheric carbon stocks and temperatures: by 2100, atmospheric carbon is almost 80 GtC (about 6%) higher, translating into an additional 0.2°C (also ~ 6%) of warming relative to the static models. Consequently, the optimal social cost of carbon rises by as much as US\$17 per tCO₂, necessitating stronger mitigation. These results underscore the importance of explicitly representing the land biosphere reservoir when modelling land use change. Deforestation and urbanisation materially alter optimal carbon policy, highlighting the need for coordinated measures that address both industrial emissions and land management. Under an extreme warming scenario, the incremental impact of deforestation is comparable in magnitude, reinforcing its relevance. Overall, our framework provides a robust, interpretable basis for climate-economic analysis, overcoming the limitations of reduced-reservoir models and enabling the study of extreme cases.

Pattern scaling (Santer *et al.*, 1990; Tebaldi and Arblaster, 2014; Lynch *et al.*, 2017) remains the workhorse for downscaling global mean projections to gridded fields, recently entering economic analyses of spatial damages and insurance (Krusell and Smith, 2022; Cruz and Rossi-Hansberg, 2024; Desmet and Rossi-Hansberg, 2024; Kotlikoff *et al.*, 2024). Using the pattern library of Lynch *et al.* (2017), our module turns the global mean temperature path from any emulator into a gridded warming map and, when absolute temperatures matter, anchors that map to an observation-based climatology such as ERA5.⁴ Because the module keeps the three main uncertainty sources, emulator calibration, ESM pattern choice and observational baseline, explicit and separate, analysts can see immediately how each factor propagates to regional damages. Pattern scaling, our computationally efficient bridge from global to local climate, unveils substantial geographic and methodological uncertainty. Using the pattern library of Lynch *et al.* (2017), which is derived from the forty-one global climate models in the Coupled

³ Afforestation, reforestation, and soil-carbon enhancement could store hundreds of petagrams of carbon (Fuss *et al.*, 2018; Minx *et al.*, 2018; Nemet *et al.*, 2018). Sophisticated IAMs such as GCAM (Calvin *et al.*, 2014) and IMAGE (Doelman *et al.*, 2020) model these processes, but at a high computational and transparency cost. Earlier attempts to link DICE with forestry models (Sohngen and Sedjo, 2006) lacked a land box in the climate block. By embedding both static and dynamic land reservoirs inside a lightweight CE, we bridge this gap while retaining the tractability of cost–benefit IAMs.

⁴ ERA5 is the European Centre for Medium-Range Weather Forecasts (ECMWF) fifth-generation re-analysis, an hourly, global dataset that merges observations with a modern forecast model; see <https://www.ecmwf.int/en/forecasts/dataset/ecmwf-reanalysis-v5>.

Model Intercomparison Project, Phase 5 (CMIP5), shows that land areas typically heat about 50% faster than the planet as a whole, although the exact amplification varies markedly between models and regions. Anchoring a given pattern to different present-day temperature maps, for example, the ERA5 re-analysis versus the climatology of an individual model, can further shift the projected 2100 regional means by up to 3°C.⁵ Passing these alternative temperature fields through a standard hump-shaped damage function can therefore flip subregions from ‘relative winners’ to ‘relative losers’ and vice versa.

Section 1 defines the class of carbon cycle emulators that we study, and Section 2 details the constrained optimisation routine that calibrates any CCE for use in a full climate emulator. Section 3 examines how the emulator design and fitting choices propagate through a dynamic economic model and shape the resulting optimal climate policies. Section 4 formalises a plug-and-play pattern-scaling module, quantifies how different ESMs and observational baselines affect regional warming and absolute temperatures, and demonstrates, through a spatially resolved damage function, how these uncertainties translate into heterogeneous economic impacts. Section 5 concludes. All source code is openly available at <https://github.com/ClimateChangeEcon/Building-Interpretable-Climate-Emulators-forEconomics>.

1. Carbon Cycle

In this section, we present the most general formulation of the linear box models for the carbon cycle used in this paper. The carbon cycle governs atmospheric CO₂ concentrations and, through interactions among several carbon reservoirs, modulates global temperature, as illustrated in Figure 1. Our objective is to demonstrate to quantitative economists how transparent CCEs can be selected, or even custom-designed, to address specific research questions within a given IAM, such as those concerning deforestation or reforestation. To that end, we analyse three representative CCE specifications.

We begin in Section 1.1 with a formal description of a multi-reservoir, linear carbon cycle comprising three reservoir classes: atmosphere (A), ocean (O) and land biosphere (L). We first analyse a three-reservoir, serial configuration (denoted as 3SR), in which carbon moves sequentially from the atmosphere through two vertically stacked ocean reservoirs, that is, the upper ocean (O₁) and deep ocean (O₂). Next, we introduce a novel four-reservoir, parallel configuration (denoted as 4PR), where atmospheric carbon is partitioned into concurrent flows towards the land biosphere and the upper ocean. We then extend the 4PR configuration to a dynamic variant (denoted as 4PR-X) by incorporating a time-dependent operator that captures shifts in the equilibrium state of the carbon cycle, most notably changes in the land biosphere storage capacity, thereby enabling the simulation of scenarios with diminished carbon uptake, such as those induced by deforestation. Section 1.2 outlines potential challenges in calibrating these models and motivates the fitting methodology detailed in Section 2. After calibration, we evaluate each CCE in two settings: (i) within a dynamic economic model analysed in Sections 3 of the main text, and (ii) in a complementary suite of stand-alone climate-science experiments reported in Online Appendix A.

⁵ CMIP5 is the international multi-model ensemble that underpinned the Intergovernmental Panel on Climate Change’s (IPCC) ‘Fifth Assessment Report’; it provides internally consistent past-to-future climate simulations for dozens of earth system models. See <https://wcrp-cmip.org/cmip-phases/cmip5> for CMIP5 and <https://www.ipcc.ch> for the IPCC.

1.1. Multi-Reservoir Model

Let $\mathbf{m}_t \in \mathbb{R}^n$ be the vector of carbon masses held in n reservoirs at discrete times $t = 0, \dots, T - 1$ and let $\mathbf{A} \in \mathbb{R}^{n \times n}$ denote the linear operator whose entry $\mathbf{A}_{ij}$ quantifies the flux *from* reservoir j *to* reservoir i .⁶ By choosing which entries of $\mathbf{A}$ are non-zero, the modeller stipulates the routes along which carbon is allowed to circulate. Figure 1 depicts the permissible CO₂ flow patterns of the four-reservoir carbon cycle emulator (4PR, $n = 4$), comprising the atmosphere, upper ocean, deep ocean and land biosphere. In this model, CO₂ is exchanged bidirectionally between the atmosphere and the upper ocean, between the atmosphere and the land biosphere and between the upper and deep ocean layers.⁷

Denote $\mathbf{e}_t \in \mathbb{R}^n$ as the external carbon emissions at discrete time t (positive values correspond to inputs to the system). The evolution of the reservoir mass vector $\mathbf{m}_t \in \mathbb{R}^n$ is governed by the linear first-order difference equation

$$\mathbf{m}_{t+1} - \mathbf{m}_t = \mathbf{A} \mathbf{m}_t + \mathbf{e}_t, \quad t = 0, 1, \dots \quad (1)$$

with known initial condition $\mathbf{m}_0$. Unless stated otherwise, we adopt an annual time step, so t indexes calendar years. Emissions may arise from both anthropogenic activity and natural processes.⁸ We summarise below the structural properties of the operator $\mathbf{A}$ that are required for the calibration procedure in Section 2.1.

- The operator $\mathbf{A}$ possesses real eigenvalues $\lambda_i(\mathbf{A})$ confined to the interval $(-1, 0]$. For each non-zero eigenvalue, we define the associated dynamic timescale as $\tau_i = 1/|\lambda_i|$.⁹ Accordingly, the admissible timescales for the linear carbon cycle model satisfy $\tau_i \in (1, \infty)$.¹⁰
- The operator $\mathbf{A}$ is subject to the equilibrium (steady-state) constraint

$$\mathbf{A} \tilde{\mathbf{m}} = \mathbf{0}, \quad (2)$$

where $\tilde{\mathbf{m}}$ denotes the vector of equilibrium carbon masses and is, up to normalisation, the eigenvector associated with the zero eigenvalue of $\mathbf{A}$.

- The principle of mass conservation for $\mathbf{A}$ is upheld by ensuring that for all t the following holds:¹¹

$$\mathbf{1}^\top (\mathbf{m}_{t+1} - \mathbf{m}_t) = \mathbf{1}^\top \mathbf{e}_t \iff \mathbf{1}^\top \mathbf{A} = \mathbf{0}. \quad (3)$$

- For each pair of reservoirs i and j , a carbon transfer pathway exists if $\mathbf{A}_{ij} \neq 0$, and no pathway exists if $\mathbf{A}_{ij} = 0$ (see Figure 2).

⁶ Throughout, subscripts denote time in time-dependent variables, whereas superscripts index vector or matrix components. Thus, $\mathbf{m}_t^1$ and $\mathbf{m}_t^A$ represent the reservoir mass of component 1 and reservoir A, respectively, at time t . For equilibrium quantities, only subscripts are used to label the components; for example, $\tilde{\mathbf{m}}_A$ is the equilibrium carbon mass in the atmosphere.

⁷ The three-reservoir, serial scheme (3SR, $n = 3$) is obtained by suppressing the A–L exchange pathway.

⁸ Examples include fossil fuel combustion, land use change, volcanic eruptions, and wildfires.

⁹ An explicit form of the operator is provided in [Online Appendix C](#).

¹⁰ The eigenvalue $\lambda_i = 0$ represents the equilibrium mode and implies an infinite timescale; this mode is omitted from the reported set.

¹¹ The principle of mass conservation states that the total mass in a closed system remains constant over time, meaning that mass can only be redistributed, but is neither created nor destroyed. In the context of the carbon cycle, this principle ensures that the total carbon mass across all reservoirs remains constant, regardless of internal exchanges among reservoirs.

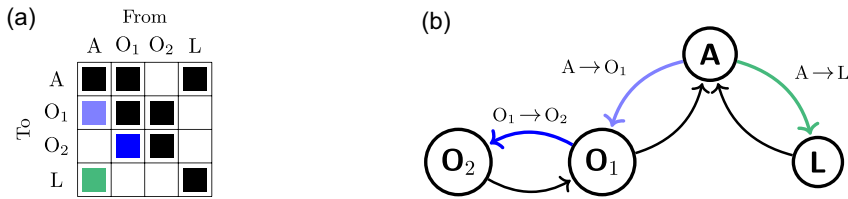


Fig. 2. Panel (a) Depicts the Non-Zero Pattern of the Operator Matrix $\mathbf{A}$ for the Four-Reservoir Carbon Cycle Emulator (4PR), Whose Reservoirs Are the Atmosphere (A), Upper Ocean (O₁), Deep Ocean (O₂) and Land Biosphere (L). Black squares indicate fixed, non-zero coefficients, while coloured squares mark free parameters that are calibrated, see equation (4). Matrix rows ('To') correspond to recipient reservoirs and columns ('From') to donor reservoirs. In panel (b), we show the directed graph of the same model, which corresponds to the possible transfer pathways of carbon. Nodes denote reservoirs, and arrows represent admissible bidirectional carbon fluxes; coloured arrows map one-to-one onto the free parameters highlighted in panel (a). For example, the flux O₁ → O₂ is entry $\mathbf{A}_{3,2}$. Eliminating the land–atmosphere exchange by setting $\mathbf{A}_{4,1} = 0$ reduces the system to the three-reservoir (3SR) model.

- We denote $\mathcal{I} := \{(i, j) : \mathbf{A}_{ij} \neq 0, i > j\}$ as the strictly lower triangular non-zero indices of $\mathbf{A}$. For all model configurations, the upper triangular values of the operator satisfying both (2) and (3) are given by:

$$\mathbf{A}_{ji} = \mathbf{A}_{ij} \cdot \frac{\tilde{\mathbf{m}}_j}{\tilde{\mathbf{m}}_i} \quad \forall (i, j) \in \mathcal{I}, \text{ and } \mathbf{A}_{ii} = - \sum_{j=1, j \neq i}^p \mathbf{A}_{ji}. \quad (4)$$

The parameters defining the operator, denoted $\mathbf{A}[\mathbf{a}, \tilde{\mathbf{m}}]$, are the strictly lower triangular non-zero elements $\mathbf{a} := \{\mathbf{A}_{ij} : (i, j) \in \mathcal{I}\}$, along with the equilibrium masses $\tilde{\mathbf{m}}$.

- The set of admissible parameters includes those that satisfy $-1 < \lambda_i(\mathbf{A}[\mathbf{a}, \tilde{\mathbf{m}}]) \leq 0$. For a predefined sequence of emissions $\mathbf{E} := (\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_T)$ of length T , the resulting carbon cycle simulation is denoted as

$$\mathbf{M}[\mathbf{a}, \tilde{\mathbf{m}}] := (\mathbf{m}_1, \mathbf{m}_2, \dots, \mathbf{m}_T), \quad (5)$$

where each $\mathbf{m}_t$ is defined according to (1). For example, the carbon mass in the deep-ocean reservoir (O₂) at time t is denoted as $\mathbf{M}[\mathbf{a}, \tilde{\mathbf{m}}]_t^{\text{O}_2} = \mathbf{m}_t^{\text{O}_2}$, consistent with the definition in Equation (5).

Having introduced the general framework, we now focus on the four-reservoir (4PR) model. The system of equations described in (1) for the four-reservoir model can be explicitly expressed

as follows in (6):

$$\begin{aligned}
 \mathbf{m}_{t+1}^A &= (1 - \mathbf{A}_{2,1} - \mathbf{A}_{4,1}) \cdot \mathbf{m}_t^A + \mathbf{A}_{2,1} \cdot \frac{\tilde{\mathbf{m}}_A}{\tilde{\mathbf{m}}_{O_1}} \cdot \mathbf{m}_t^{O_1} + \mathbf{A}_{4,1} \cdot \frac{\tilde{\mathbf{m}}_A}{\tilde{\mathbf{m}}_L} \cdot \mathbf{m}_t^L \\
 \mathbf{m}_{t+1}^{O_1} &= \mathbf{A}_{2,1} \cdot \mathbf{m}_t^A + \left(1 - \mathbf{A}_{2,1} \cdot \frac{\tilde{\mathbf{m}}_A}{\tilde{\mathbf{m}}_{O_1}} - \mathbf{A}_{3,2}\right) \cdot \mathbf{m}_t^{O_1} + \mathbf{A}_{3,2} \cdot \frac{\tilde{\mathbf{m}}_{O_1}}{\tilde{\mathbf{m}}_{O_2}} \cdot \mathbf{m}_t^{O_2} \\
 \mathbf{m}_{t+1}^{O_2} &= \mathbf{A}_{3,2} \cdot \mathbf{m}_t^{O_1} + \left(1 - \mathbf{A}_{3,2} \cdot \frac{\tilde{\mathbf{m}}_{O_1}}{\tilde{\mathbf{m}}_{O_2}}\right) \cdot \mathbf{m}_t^{O_2} \\
 \mathbf{m}_{t+1}^L &= \mathbf{A}_{4,1} \cdot \mathbf{m}_t^A + \left(1 - \mathbf{A}_{4,1} \cdot \frac{\tilde{\mathbf{m}}_A}{\tilde{\mathbf{m}}_L}\right) \cdot \mathbf{m}_t^L.
 \end{aligned} \tag{6}$$

Note that the 3SR carbon cycle can be derived from the same set of equations by setting $\mathbf{A}_{4,1} = 0$.

1.1.1. Time-dependent land capacity

In principle, the operator $\mathbf{A}$ may be either time-invariant (i.e., constant) or time-dependent. As defined in (4), it is time-invariant under the assumptions of (i) a fixed equilibrium partitioning of the total carbon mass across reservoirs, expressed by the ratios $\tilde{\mathbf{m}}_j/\tilde{\mathbf{m}}_i$, and (ii) constant inter-reservoir exchange coefficients $\mathbf{A}_{ij}$. However, land use-related emissions, primarily permanent deforestation and agricultural expansion, alter the storage capacity of the land biosphere (Hurt et al., 2011; Meinshausen et al., 2011). Within the box model framework, the equilibrium carbon mass of the land reservoir, therefore, cannot remain constant.

When changes in the equilibrium mass of the land biosphere $\tilde{\mathbf{m}}_L$ are taken into account over time, the operator $\mathbf{A}$ becomes time-dependent and we denote it as $\mathbf{A}_t$. Given land use change emissions $\mathbf{e}_t^L$ at time t , and assuming that a fraction r of these emissions results from deforestation, the equilibrium mass of the land biosphere in the next time step is

$$\tilde{\mathbf{m}}_{t+1}^L = \tilde{\mathbf{m}}_t^L - r \cdot \mathbf{e}_t^L. \tag{7}$$

This modification in (7) yields the time-dependent version of the operator in (4). In our simplified setting we adopt a one-to-one correspondence ($r = 1$): every unit of land use emission decreases the equilibrium mass $\tilde{\mathbf{m}}_L$ by the same amount, so that all land use emissions directly reduce the land biosphere reservoir. The equilibrium mass in the box model nonetheless only approximates the true carbon stored on land.¹² Therefore, a one-to-one correspondence between real-world carbon emissions from land use change and the change in equilibrium land biosphere mass in the emulator may not be guaranteed. The proportionality factor r could be different from 1.

The sensitivities of atmospheric and land biosphere reservoir masses to changes in land biosphere equilibrium mass can be expressed as follows:

$$\frac{\partial \mathbf{m}_{t+1}^A}{\partial \tilde{\mathbf{m}}_t^L} \propto -\frac{1}{(\tilde{\mathbf{m}}_t^L)^2}, \quad \frac{\partial \mathbf{m}_{t+1}^L}{\partial \tilde{\mathbf{m}}_t^L} \propto \frac{1}{(\tilde{\mathbf{m}}_t^L)^2}. \tag{8}$$

Hence, it follows from Equation (8) that a decrease in the land biosphere equilibrium mass will (i) increase the carbon remaining in the atmosphere, and (ii) reduce the carbon retained in the land biosphere. Simulation results confirming this behaviour are presented in [Online Appendix A.2](#). Although the time-stepping formulation in (1) remains linear, the above analysis indicates a

¹² By contrast, the atmospheric carbon stock in the model matches the historical value of 589 GtC in the year 1750; see Table 1.

potentially non-linear sensitivity of carbon mass exchanges with respect to changes in the land biosphere equilibrium mass.

1.2. Challenges of Degrees of Freedom

A central challenge in constructing a climate emulator is to determine a parameter set that allows the reduced model to replicate the carbon cycle behaviour of more detailed earth system models. The usual remedy is to solve an optimisation problem that minimises the discrepancy between the emulator's output and a benchmark simulation. In practice, however, this calibration task is often ill-posed: different parameter vectors can generate virtually identical carbon flux trajectories, even though many of those vectors are physically implausible.

The indeterminacy originates partly from (4), which shows that only the ratios of the equilibrium masses enter the operator $\mathbf{A}$. Hence, for any scalar $c > 0$, the replacement $\tilde{\mathbf{m}} \mapsto c \tilde{\mathbf{m}}$ leaves the model response unchanged. In Section 2 we resolve this degeneracy by introducing a physics-informed calibration scheme that (i) enforces additional physical constraints and (ii) exploits the emulator's linear structure to construct a weighted operator capable of capturing the extreme dynamical regimes present across the admissible model configurations.

2. A Calibrated Climate Emulator

In this section, we introduce a systematic calibration procedure that turns a box type CCE, configured for a specific integrated assessment purpose (see Section 1), into a climate data-constrained component ready for quantitative modelling. Furthermore, we show how the calibrated CCE can be coupled with a temperature module to form a full climate emulator that can be embedded in quantitative IAMs (Figure 1). Recall that, throughout this paper, the term emulator refers to this combined carbon cycle and temperature model.

Section 2.1 presents a generic fitting framework for the linear box model with n reservoirs (see Section 1), encompassing data selection, parameter estimation and hyperparameter tuning. As an example, we utilise the pulse decay database of Joos *et al.* (2013), which compiles simulations of instantaneous carbon release from a suite of state-of-the-art climate models worldwide; these trajectories serve as our fitting target. Individual simulations differ, among others, in the climate model used, the amplitude of the pulse, or the background conditions into which the pulse is released. In our work below, we use the so-called multi-model mean (average over more than ten individual models) of the simulated decay of a 100 GtC carbon pulse under pre-industrial (PI) conditions, corresponding to the year 1750. To illustrate that other choices are possible, we provide in Online Appendix B a corresponding illustration of our method for present-day (PD) background conditions, representative of the year 2010. Any pulse decay dataset, not only that of Joos *et al.* (2013), can be used with our toolbox to calibrate the CCE. The selection should match the intended application: for example, whether average or extreme decay behaviour is required (Folini *et al.*, 2025), and which background state (PI, PD, or another) is most relevant.

Next, in Section 2.2, we outline the temperature model and its parameterisation, which is primarily based on the work of Geoffroy *et al.* (2013a,b), with slight adjustments to account for external radiative forcing factors.¹³

¹³ Radiative forcing factors are constituents or processes that disturb the earth's energy balance by altering the net flux of incoming solar or outgoing longwave radiation. Typical examples include long-lived greenhouse gases (e.g.,

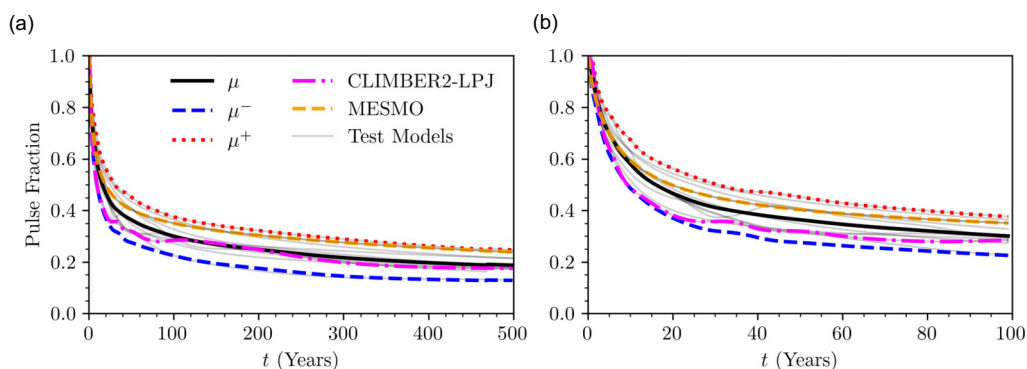


Fig. 3. Simulated Atmospheric CO_2 Decay After a 100 GtC Pulse Under PI Equilibrium Conditions with Experimental Data from Joos *et al.* (2013). In panels (a) and (b), we show the decay trajectories of different models for 0 – 500 and 0 – 100 years, respectively. Shown is the simulated pulse decay across various test models (thin grey lines), including ESMs and EMICs. Shown also is the multi-model mean (μ , thick black line), with μ^+ (thick dashed line) and μ^- (thick dotted line) representing 2 SDs above and below the multi-model mean, respectively. CLIMBER2-LPJ (Sitch *et al.*, 2005) and MESMO (Matsumoto *et al.*, 2021) illustrate two models with fast and slow decay trajectories, respectively, representing extremes relative to the multi-model mean.

To build a bridge to practical applications, [Online Appendix C](#) provides a concise guide for climate economists, explaining how to deploy each emulator presented in this paper to various targets, including PI, PD, multi-model mean and extreme scenarios.

2.1. Carbon Cycle Fitting Procedure

To fit the CCE's free parameters to the data shown in Figure 2(a), we apply a constrained optimisation procedure based on the pulse decay dynamics of Joos *et al.* (2013).¹⁴ The calibration proceeds in two steps, detailed in Sections 2.1.1 and 2.1.2. First, we adjust the parameters so that the model reproduces the mean atmospheric pulse decay trajectory derived from benchmark simulations. Second, we rescale this calibrated model to capture extreme carbon cycle responses.

The simulation benchmarks are the atmospheric CO_2 decay trajectories generated by several carbon cycle models after a 100 GtC pulse is injected into an initially equilibrated system (i.e., no net carbon flux between reservoirs).¹⁵ Figure 3 shows the benchmark data: the multi-model mean, denoted μ , together with μ^+ and μ^- , which lie 2 SDs above and below μ , respectively, as reported by Joos *et al.* (2013). For reference, two individual models, CLIMBER2-LPJ and MESMO, illustrate very fast and slow pulse decay responses.

2.1.1. Multi-model mean calibration

Let $\mathbf{y}^\mu \in \mathbb{R}^T$ denote the atmospheric decay trajectory for the μ -benchmark, that is, the multi-model mean, for T years after the introduction of the 100 GtC pulse.¹⁶ The fit error, which

CO_2 , CH_4 , N_2O), short-lived species such as aerosols or tropospheric ozone, surface-albedo changes from land use, and variations in solar irradiance or volcanic emissions.

¹⁴ A detailed discussion of why we chose this dataset to fit the CCE is provided in Folini *et al.* (2025).

¹⁵ See <https://climatehomes.unibe.ch/~joos/IRF/Intercomparison/results.html> for further details.

¹⁶ We use the superscript μ throughout for quantities derived from the benchmark dataset, here the multi-model mean. Subsequent sections introduce analogous notation for other datasets.

is measured here based on the deviation between the emulated atmospheric masses and the multi-model mean, that is, our so-called μ -benchmark, is defined as follows:¹⁷

$$\mathcal{L}(\mathbf{a}^\mu, \tilde{\mathbf{m}}^\mu) := \frac{1}{T} \left\| \mathbf{M}[\mathbf{a}, \tilde{\mathbf{m}}]^A - \mathbf{y}^\mu \right\|_2. \quad (9)$$

For example, in the 4PR model, $\mathbf{a}^\mu$ is a three-element vector representing carbon mass transfer rates between different reservoirs (see Figure 2 for details). Likewise, $\tilde{\mathbf{m}}^\mu$ is a four-element vector containing the equilibrium carbon mass of each reservoir. The atmospheric equilibrium mass is fixed to the estimated pre-industrial value 589 GtC (Ciais *et al.*, 2014; see Online Appendix B for the present-day calibration).

Since the benchmark data only contains atmospheric carbon masses, the models will be over-parameterised relative to the objective. Consider the 4PR model with atmospheric equilibrium fixed: six parameters are fitted to a single output $\mathbf{m}_t^A$, while the other carbon reservoirs remain free to take on any value. This is problematic because it results in a highly under-constrained model, where different parameter sets can yield the same atmospheric reservoir mass, but vastly different carbon distributions across the other reservoirs. To address this, we introduce three penalty terms, q_1 , q_2 and q_3 , which penalise deviations from physically motivated behaviour observed in comprehensive carbon cycle simulations. For the models considered in this work, we have selected penalty functions to target deviations in dynamic timescales (q_1), equilibrium mass variability (q_2) and reservoir absorption ratios (q_3).¹⁸ Given the non-negative scalar hyperparameters ρ_1 , ρ_2 and ρ_3 , each corresponding to the respective penalty functions, the μ -benchmark fitted parameters are

$$\left\{ \mathbf{a}^\mu, \tilde{\mathbf{m}}^\mu \right\} = \underset{\{\mathbf{a}, \tilde{\mathbf{m}}\}}{\operatorname{argmin}} \left\{ \mathcal{L}(\mathbf{a}, \tilde{\mathbf{m}}) + \rho_1 q_1(\mathbf{a}, \tilde{\mathbf{m}}) + \rho_2 q_2(\tilde{\mathbf{m}}) + \rho_3 q_3(\mathbf{a}, \tilde{\mathbf{m}}) \right\}. \quad (10)$$

In our tests, we use $T = 250$, aligning with the timescales associated with the carbon exchange between the atmosphere and the earth's surface, ranging from decades to centuries. Below, we provide a detailed description of the penalty functions.

Dynamic timescales (q_1). In addition to surface–atmosphere carbon exchange, we also account for the deep-ocean carbon cycle, which operates on centennial to millennial timescales (Ciais *et al.*, 2014). As such, we penalise model parameters $\mathbf{a}$ and $\tilde{\mathbf{m}}$ that yield an operator $\mathbf{A}$ with short dynamic timescales. Each eigenvalue $\lambda_i \leq 0$ of $\mathbf{A}$ defines an exponential decay mode with dynamic timescale $\tau_i = 1/|\lambda_i|$ (see, e.g., Åström and Murray, 2021, ch.5); smaller $|\lambda_i|$ therefore implies slower decay, and thus larger values of τ_i . To enforce this characteristic in the model, we propose the following penalty function:

$$q_1(\mathbf{a}, \tilde{\mathbf{m}}) := -\frac{1}{n} \operatorname{tr}(\mathbf{A}[\mathbf{a}, \tilde{\mathbf{m}}]) = -\frac{1}{n} \sum_{i=1}^n \lambda_i = \frac{1}{n} \sum_{i=1}^n |\lambda_i|. \quad (11)$$

¹⁷ Note that, whereas Folini *et al.* (2025) employed the maximum (ℓ_∞) norm, we use the Euclidean (ℓ_2) norm in the present work; both choices are equally defensible. Their choice of the maximum norm was motivated by obtaining a better fit to the early part of the pulse decay, whereas the ℓ_2 -norm places relatively greater weight on the tail of the decay. Our fitting period spans 250 years, compared with the 100 years used in that study.

¹⁸ Additional penalty terms may be introduced to capture further physical aspects of the carbon cycle, but they must be chosen carefully to avoid redundancy. Each penalty should target a specific feature, such as a statistical regularity or a dynamical constraint, identified from observations or large-scale simulations.

Notice that this penalty function in Equation (11) is strictly non-negative because every $\lambda_i \leq 0$. When included in the objective, this penalty function suppresses the average eigenvalue magnitude of the operator, promoting slower system dynamics and longer dynamic timescales.

Equilibrium mass variability (q_2). The optimisation problem stated in (10) is non-convex, as is evident from (4), which depends only on the ratios of $\tilde{\mathbf{m}}$, making any positive scalar multiple of $\tilde{\mathbf{m}}$ equally valid in determining $\mathbf{A}$. Consequently, the fitted parameters $\tilde{\mathbf{m}}$ (and thus $\mathbf{a}$) may vary substantially across equally valid solutions. To enforce consistency with observational constraints, we add a penalty term for squared deviations between the model's equilibrium reservoir masses and their target values, where the targets here are the published estimates of PI (or, alternatively, PD) carbon stocks. Let $\tilde{\mathbf{m}}^*$ denote these estimated equilibrium masses. We penalise the relative difference between $\tilde{\mathbf{m}}$ and a reference equilibrium vector $\tilde{\mathbf{m}}^*$ using the penalty function:

$$q_2(\tilde{\mathbf{m}}) := \frac{1}{n} \left\| (\tilde{\mathbf{m}} - \tilde{\mathbf{m}}^*) \oslash \tilde{\mathbf{m}}^* \right\|_2, \quad (12)$$

where $\oslash$ denotes element-wise division. In our tests, $\tilde{\mathbf{m}}^*$ is defined based on the PI estimates from Ciais *et al.* (2014), where the equilibrium masses of the reservoirs for the atmosphere, upper ocean, lower ocean, and land biosphere are 589, 900, 37100 and 550 GtC, respectively. We emphasise that the values of $\tilde{\mathbf{m}}^*$ do not represent the absolute total equilibrium masses of carbon in each reservoir. Instead, these values estimate the carbon masses most actively involved in the carbon cycle dynamics. In contrast, carbon stored in fossil fuel reserves, permafrost, and deep soil organic matter is generally sequestered on timescales ranging from centuries to millennia, and is therefore excluded from equilibrium carbon mass estimates.¹⁹

Reservoir absorption ratios (q_3). Under a 100 GtC pulse, the 4PR model may achieve a low fit error while exhibiting negligible flux to the land biosphere, effectively replicating the behaviour of the 3SR model (see Section 2 for further details). This observation is specific to the 4PR configuration, where parallel carbon transfer paths—from the atmosphere to either the ocean or the land biosphere—allow for an arbitrary partitioning of carbon. To better reflect the behaviour of more complex earth system models (ESMs), we will aim to mimic the ratio of cumulative ocean to land biosphere fluxes resulting from pulse emissions. Let η denote the target ratio of cumulative ocean to land biosphere carbon masses at time t^{ref} ; we penalise deviations from this ratio using the following penalty function:

$$q_3(\mathbf{a}, \tilde{\mathbf{m}}) := \left\| \frac{\mathbf{M}[\mathbf{a}, \tilde{\mathbf{m}}]_{t^{\text{ref}}}^{\text{O}}}{\mathbf{M}[\mathbf{a}, \tilde{\mathbf{m}}]_{t^{\text{ref}}}^{\text{L}}} - \eta \right\|_2. \quad (13)$$

In Equation (13), $\mathbf{M}[\mathbf{a}, \tilde{\mathbf{m}}]_{t^{\text{ref}}}^{\text{O}}$ denotes the total mass of all ocean reservoirs (O_1 and O_2), while $\mathbf{M}[\mathbf{a}, \tilde{\mathbf{m}}]_{t^{\text{ref}}}^{\text{L}}$ denotes the total mass of all land biosphere reservoirs. Following the findings of Joos *et al.* (2013), we set $\eta = 1$, corresponding to an equal distribution of 30 GtC between ocean and land biosphere reservoirs at $t^{\text{ref}} = 20$ years after a 100 GtC atmospheric pulse.

2.1.2. Calibration for extreme scenarios

As demonstrated in Figure 3, different ESMs exhibit varying 100 GtC pulse decay trajectories. In this section, we aim to capture the extrema, defined as 2 SDs above and below the mean atmospheric response across the trajectories of different ESMs, labelled as the μ^+ -benchmark

¹⁹ However, carbon stored in permafrost can rapidly re-enter the active carbon cycle when disturbed, particularly through climate-induced thawing. While this represents a potentially significant perturbation to the carbon cycle, it is not accounted for in the present analysis and warrants further investigation.

and μ^- -benchmark, respectively. We emphasise that the benchmarks used here (and also in the previous section) do not correspond to any specific ESM, but represent plausible upper and lower bounds for atmospheric carbon masses.

A simple way to calibrate the model for the different extrema is to rescale the parameters $\mathbf{a}^\mu$ obtained in (10). This formulation allows IAM researchers to interpolate smoothly between the mean response and its extremes, eliminating the need to recalibrate the CCE each time a more or less extreme scenario is examined.

Given the pulse decay trajectories $\mathbf{y}^{\mu^+}, \mathbf{y}^{\mu^-} \in \mathbb{R}^T$, we define extreme scaling factors as

$$\begin{aligned} c^{\mu^+} &= \operatorname{argmin}_c \left\{ \frac{1}{T} \left\| \mathbf{M}[c \cdot \mathbf{a}^\mu, \tilde{\mathbf{m}}^\mu]_{\mathbf{A}} - \mathbf{y}^{\mu^+} \right\|_2 \right\}, \text{ and} \\ c^{\mu^-} &= \operatorname{argmin}_c \left\{ \frac{1}{T} \left\| \mathbf{M}[c \cdot \mathbf{a}^\mu, \tilde{\mathbf{m}}^\mu]_{\mathbf{A}} - \mathbf{y}^{\mu^-} \right\|_2 \right\}. \end{aligned} \quad (14)$$

For $\mathbf{A}^\mu := \mathbf{A}[\mathbf{a}^\mu, \tilde{\mathbf{m}}^\mu]$, we define the respective extrema operators as $\mathbf{A}^{\mu^+} := c^{\mu^+} \cdot \mathbf{A}^\mu$ and $\mathbf{A}^{\mu^-} := c^{\mu^-} \cdot \mathbf{A}^\mu$. In our test, we keep T consistent with the value used in (10). Note that $\mathbf{A}^{\mu^+}$ and $\mathbf{A}^{\mu^-}$ are scaled versions of $\mathbf{A}^\mu$, with all eigenvalues multiplied by c^{μ^+} and c^{μ^-} , respectively.²⁰ Scaling the eigenvalues also shifts the range of dynamic timescales. The solutions in (14) will satisfy $c^{\mu^+} < 1 < c^{\mu^-}$. Due to the linearity of the carbon cycle model, we can formulate a parameterised weighted operator that satisfies the conditions outlined in Section 1.1, including the equilibrium condition and mass conservation. Given $\alpha \in [-1, 1]$, we write the weighted operator as

$$\mathbf{A}^\alpha := \begin{cases} (1 - \alpha)\mathbf{A}^\mu + \alpha\mathbf{A}^{\mu^+} & \text{if } \alpha > 0, \\ (1 + \alpha)\mathbf{A}^\mu - \alpha\mathbf{A}^{\mu^-} & \text{if } \alpha \leq 0, \end{cases} \quad (15)$$

where, as defined in Equation (15), $\alpha = 0$ represents the mean atmospheric carbon content across ESMs. Setting $\alpha = 1$ or $\alpha = -1$ emulates ESMs with higher or lower atmospheric carbon content, respectively, corresponding to slower or faster carbon absorption from the atmosphere.

2.1.3. The choice of hyperparameters

In this section, we describe the methodology and criteria used to select the hyperparameters ρ_1 , ρ_2 and ρ_3 in (12), and report the chosen values for each model. As a general principle, smaller values for these hyperparameters are preferred to avoid the penalty terms overwhelming the model fit error in the optimisation objective (9).

We begin with ρ_2 and ρ_3 , which are associated with the penalty terms for equilibrium mass variability and reservoir absorption ratios, respectively (see Section 2.1.1). For the 4PR model configuration, our numerical results indicate that setting $\rho_2 = \rho_3 = 10^{-4}$ minimises the fit error while achieving desirable low equilibrium mass variations and reservoir absorption ratios. The results are similar for the 3SR configuration, except that q_3 has no effect due to the absence of a land biosphere; thus, we select $\rho_2 = 10^{-4}$ and $\rho_3 = 0$. For a detailed analysis and numerical results on model behaviour related to ρ_2 and ρ_3 see [Online Appendix A](#).

With ρ_2 and ρ_3 fixed, we now focus on selecting an appropriate value for ρ_1 . To justify our choice, we evaluate the model under varying values of ρ_1 , analysing (i) the minimum, average and maximum dynamical timescales, and (ii) the average absolute error in atmospheric carbon content relative to the μ -benchmark over the time horizon $50 \leq T \leq 500$ (see Figure 3). This analysis is visualised in Figure 4. For sufficiently small values of ρ_1 (those for which the penalty

²⁰ Specifically, $\mathbf{A}[c \cdot \mathbf{a}, \tilde{\mathbf{m}}]$ is equivalent to $c \cdot \mathbf{A}_{ij}$; see (4) for details.

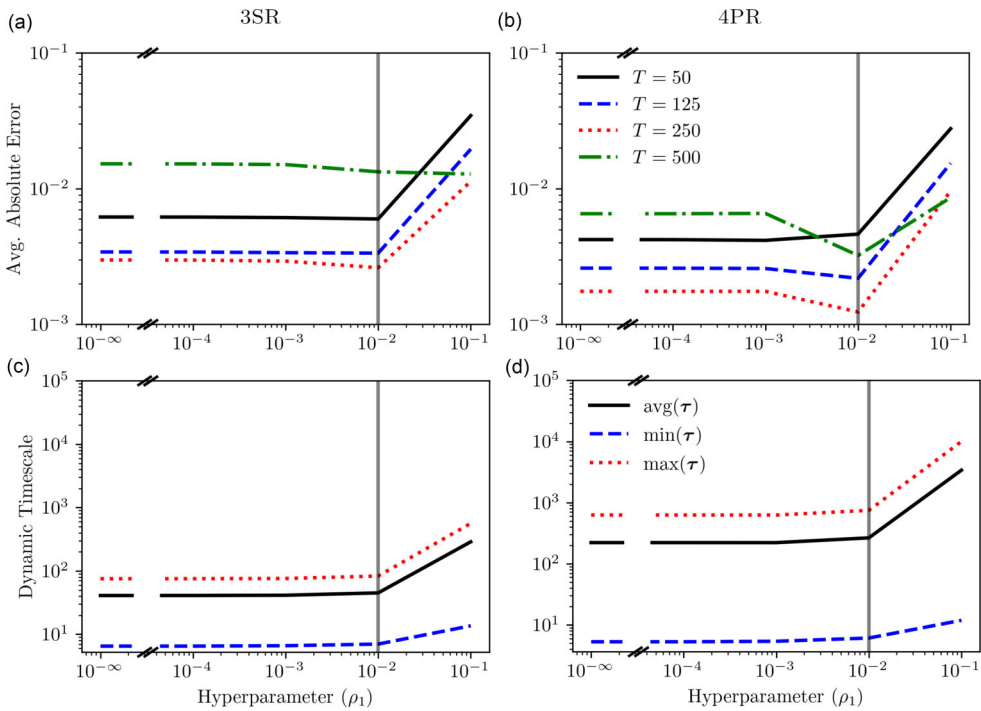


Fig. 4. Panels (a) and (b) Show the Average Absolute Error ($\|\mathbf{m}_A - \mathbf{y}^\mu\|_1/T$), and Panels (c) and (d) Show the Dynamic Timescales τ for the 3SR (Panels (a) and (c)) and 4PR (Panels (b) and (d)) Models Across Varying ρ_1 , with $\rho_2 = \rho_3 = 10^{-4}$ Fixed. The average absolute error represents the mean absolute difference between the emulator simulation and the μ -benchmark over a time span of T years following a 100 GtC carbon pulse. The vertical line indicates the selected value of $\rho_1 = 10^{-2}$.

function does not overwhelm the fit error), the 3SR model exhibits minimal sensitivity in terms of both average absolute error and dynamical timescales, primarily due to its reduced number of parameters compared to the 4PR model. In contrast, for the 4PR model, increasing ρ_1 leads to a notable reduction in average absolute error, particularly at longer simulation times T . In both model configurations, increasing ρ_1 results in a relatively small increase in the dynamic timescales. Importantly, selecting relatively small hyperparameter values, even when they have little effect on the estimated parameters, can significantly improve the numerical stability and time to solution of the optimisation problem in (10). This observation is well established in the numerical optimisation literature (see, for example, Nocedal and Wright, 2006, for a detailed discussion of conditioning and the effects of regularisation). We proceed with selecting $\rho_1 = 10^{-2}$ for both the 3SR and 4PR model configurations.

2.1.4. Fitted model parameter

Given the hyperparameters outlined in the previous section, we now perform the multi-model mean and extrema calibrations, as discussed in Sections 2.1.1 and 2.1.2, respectively. Table 1 presents the resulting calibrated model parameters $\mathbf{a}^\mu$ and $\hat{\mathbf{m}}^\mu$, obtained as solutions to (12), along

Table 1. The Calibrated Parameter Values Obtained Using the μ -Benchmark.

Fitted parameter values											
	$a_{A \rightarrow O_1}^{\mu}$	$a_{O_1 \rightarrow O_2}^{\mu}$	$a_{A \rightarrow L}^{\mu}$	$\bar{m}_A^{\mu}$	$\bar{m}_{O_1}^{\mu}$	$\bar{m}_{O_2}^{\mu}$	$\bar{m}_L^{\mu}$	$c^{\mu+}$	$c^{\mu-}$	m_{ref}^O/m_{ref}^L	$\tau_1/\tau_2/\tau_3$
Lower	10^{-6}	10^{-6}	10^{-6}	589	10^{-6}	10^{-6}	10^{-6}	10^{-6}	1	—	7/83/—
3SR	0.0769	0.0109	—	589	752	1,289	—	0.4746	2.4559	—	—
4PR	0.0208	0.0025	0.0613	589	1,078	37,220	387	0.4701	2.4074	0.7150	6/42/748
Upper	0.3	0.3	0.3	589	1,800	74,200	1,100	1	5	—	—

Notes: The table is based on the PI pulse decay experiments reported by Joos *et al.* (2013). For calibration using PD pulse decay data, see Online Appendix B. The table also lists the lower and upper search bounds for the optimisation problems in (10) and (14), specifying the allowable range for each parameter (see Sections 2.1 and 2.2 for details on model calibration and the fitting procedure). In addition, it includes the ratio of ocean to land biosphere pulse absorption ($\bar{m}_D^L/m_{ref}^L$) and the model's dynamic timescales (τ_1, τ_2, τ_3). The ocean to land pulse absorption ratio is evaluated at $t^{ref} = 20$, see Section 2.1.1 for penalty functions. The reported values are based on the hyperparameters $\rho_1 = 10^{-2}$, $\rho_2 = 10^{-4}$, and $\rho_3 = 0$ for the 3SR model, and $\rho_1 = 10^{-2}$, $\rho_2 = 10^{-4}$, $\rho_3 = 10^{-4}$ for the 4PR model.

with the scaling factors c^{μ^+} and c^{μ^-} , which are the solutions to (14).²¹ We provide a high-level discussion of these results and their implications for model structure and dynamics below. When calibrated to either pre-industrial or present-day conditions, the three- and four-box emulators reproduce the historical and 500-year atmospheric CO₂ and global mean temperature trajectories with fitting errors below 5% and 3%, respectively (see the lower panels of Figures 15 and 24 in the [Online Appendix](#)). This level of accuracy is widely regarded as ‘fit for purpose’ in policy analysis.

The parameters are all estimated within prescribed lower and upper bounds, and provided in Table 1. The first observed difference between the 3SR and 4PR model parameters lies in the mass transfer coefficients $\mathbf{a}^{\mu}$, which are consistently smaller in the 4PR model compared to the 3SR model. This follows directly from the 4PR model structure, where atmospheric carbon is split between parallel fluxes to the ocean and land biosphere, allowing smaller individual transfer coefficients to yield the same net outflow. A second key difference between the 3SR and 4PR models concerns the range of dynamic timescales each can represent. The 3SR model, with only two non-zero eigenvalues, captures two distinct timescales, whereas the 4PR model, with three eigenvalues, resolves three timescales, including a much larger dynamic timescale associated with slower processes. A noted limitation of the 3SR model is its tendency to average the medium- and long-term dynamics, as reflected in the estimated dynamic timescale in Table 1 (also observable in Figure 4(c) and (d)). For the equilibrium masses $\hat{\mathbf{m}}^{\mu}$, the 4PR configuration yields significantly larger values than the 3SR configuration and closely aligns with those reported in Ciais *et al.* (2014)—see the discussion related to (12).

Using the calibrated parameters for the multi-model mean (i.e., calibration based on the μ -benchmark), we now turn to the extrema parameters c^{μ^+} and c^{μ^-} , which are defined according to (14), and which are calibrated using the μ^+ and μ^- benchmarks, respectively. As shown in Table 1, these values are approximately the same for both the 3SR and 4PR configurations. This is consistent with our results in [Online Appendix A.1](#), as both the 3SR and 4PR models exhibit similar atmospheric pulse decay for the μ -benchmark (since both are calibrated to this benchmark); consequently, the extrema parameters, which are based solely on atmospheric concentrations, are expected to coincide. However, this does not imply that the two models exhibit similar dynamics under general conditions. As shown in the extended results in [Online Appendix A](#) and further discussed in [Online Appendix A.1](#), the 3SR and 4PR models exhibit substantial differences in carbon distribution across individual reservoirs and in their responses to general atmospheric carbon perturbations (i.e., those not used in the calibration procedure).

2.2. Temperature Model

In this section, for completeness, we present the equations governing the evolution of temperature.²² The temperature change is given by a time-dependent radiative forcing amplitude parameter $\mathcal{F}_t$ at discrete time t , along with the following set of free parameters: the combined heat capacity of the atmosphere, upper ocean (and land biosphere) C ; the deep-ocean heat

²¹ The notation $\mathbf{a}^{\mu}$ refers to carbon transfer coefficients between the respective reservoirs, which can also be expressed in matrix form. For example, in the 4PR model: $\mathbf{A}_{2,1} := \mathbf{a}_{\text{Atmos} \rightarrow \text{O}_1}^{\mu}$, $\mathbf{A}_{3,2} := \mathbf{a}_{\text{O}_1 \rightarrow \text{O}_2}^{\mu}$, and $\mathbf{A}_{4,1} := \mathbf{a}_{\text{Atmos} \rightarrow \text{L}}^{\mu}$. In the 3SR model, the same notation applies, except that the corresponding rows and columns of $\mathbf{A}$ related to the land biosphere are omitted.

²² As discussed in the introduction, this aspect of climate emulators is extensively treated in Folini *et al.* (2025) and is therefore beyond the scope of this paper.

Table 2. *The Parameters Used for the Two-Layered Energy-Balance Model Outlined by Geoffroy et al. (2013b). It represents the global thermal characteristics of coupled atmosphere–ocean general circulation models within a two-layer energy-balance framework. For our purposes, we assume $\mathcal{F}_{4\times\text{CO}_2} = 2\mathcal{F}_{2\times\text{CO}_2}$.*

Parameters for two-layer energy-balance model					
	C (W yr m ⁻² K ⁻¹)	C_0 (W yr m ⁻² K ⁻¹)	γ (W m ⁻² K ⁻¹)	λ (W m ⁻² K ⁻¹)	$\mathcal{F}_{4\times\text{CO}_2}$ (W m ⁻²)
Multi-model mean	7.3	106	0.73	1.13	6.9
SD	1.1	62	0.18	0.31	0.9

capacity C_0 ; the heat exchange coefficient γ ; and the radiative feedback parameter λ (not to be confused with the eigenvalues discussed earlier). The summary of these parameters is presented in Table 2. Let $\mathbf{T}_t = (\mathbf{T}_t^A, \mathbf{T}_t^O)^\top \in \mathbb{R}^2$ denote global temperature at time t in a two-layer model. The first component, $\mathbf{T}_t^A$, is the temperature of the atmosphere, upper ocean and land biosphere (hereafter simply ‘atmospheric’), while the second, $\mathbf{T}_t^O$, is the temperature of the deep ocean (hereafter ‘oceanic’). The temperature dynamics are modelled as a first-order system of difference equations, expressed as

$$\mathbf{T}_{t+1} - \mathbf{T}_t = \Delta t \cdot (\mathbf{B}\mathbf{T}_t + \mathbf{b}_t), \text{ where } \mathbf{B} := \begin{bmatrix} -(\lambda + \gamma)/C & \gamma/C \\ \gamma/C_0 & -\gamma/C_0 \end{bmatrix}, \text{ and} \quad (16)$$

$$\mathbf{b}_t := \begin{bmatrix} \mathcal{F}_t/C \\ 0 \end{bmatrix}.$$

The total radiative forcing amplitude is modelled as a function of the relative CO₂ concentration, following the formulation of Houghton *et al.* (1990) and Geoffroy *et al.* (2013b). To account for additional factors, we scale the CO₂ relative forcing by a factor of κ , giving

$$\mathcal{F}_t := \kappa \cdot \frac{\mathcal{F}_{2\times\text{CO}_2}}{\ln(2)} \cdot \ln\left(\frac{\mathbf{m}_t^A}{\mathbf{m}_0^A}\right), \quad (17)$$

where $\mathbf{m}_t^A/\mathbf{m}_0^A$ in Equation (17) is the ratio of atmospheric CO₂ masses (or concentrations) at time t relative to pre-industrial (PI) levels, and $\mathcal{F}_{2\times\text{CO}_2}$ denotes the net radiative forcing associated with a doubling of atmospheric CO₂ concentration. Alternatively, explicit definitions for the temperatures of the atmosphere ($\mathbf{T}_t^A$), as well as the ocean ($\mathbf{T}_t^O$) are provided below (cf. Equation (16)):

$$\begin{aligned} \mathbf{T}_{t+1}^A &= \mathbf{T}_t^A + \frac{1}{C} \left(\kappa \cdot \frac{\mathcal{F}_{2\times\text{CO}_2}}{\ln(2)} \cdot \ln\left(\frac{\mathbf{m}_t^A}{\mathbf{m}_0^A}\right) - \gamma \cdot (\mathbf{T}_t^A - \mathbf{T}_t^O) - \lambda \cdot \mathbf{T}_t^A \right), \\ \mathbf{T}_{t+1}^O &= \mathbf{T}_t^O + \frac{\gamma}{C_0} (\mathbf{T}_t^A - \mathbf{T}_t^O). \end{aligned} \quad (18)$$

We fix $\kappa = 1.2$ in Equation (18) as the scaling factor for CO₂ forcing across all representative concentration pathway (RCP) scenarios (Folini *et al.*, 2025); this is discussed in more detail in Online Appendix A.2. For the controlled atmospheric perturbation tests in Online Appendix A.1, which are based solely on synthetic CO₂ forcing, we use $\kappa = 1$.

3. Numerical Experiments

This section systematically investigates how alternative emulator designs influence outcomes once they are embedded in a dynamic economic model. After presenting the core model in Section 3.1, we analyse (i) a business-as-usual (BAU; no-policy) case in Section 3.2, (ii) an optimal mitigation case in Section 3.3, and (iii) a hypothetical carbon capture scenario in Section 3.4.²³ For clarity, we briefly recall the three carbon cycle emulator variants introduced above, which are compared in the experiments to follow:

- (i) 3SR: a three-reservoir emulator that sequentially links the atmosphere, upper ocean, and deep ocean;
- (ii) 4PR: an extension of 3SR that adds the land biosphere as a parallel carbon sink from the atmosphere;
- (iii) 4PR-X: a 4PR variant that introduces a time-dependent operator to adjust the land biosphere equilibrium for land use-related emissions.

3.1. A Baseline Model

We embed the three CCEs (3SR, 4PR and 4PR-X) introduced in Section 1 and validated in stand-alone tests from climate science (see appendices Section A.1 and A.2) within the DICE-2016 framework (Nordhaus, 2017).²⁴ Models are calibrated to the multi-model mean and to extreme scenarios (CLIMBER2-LPJ, MESMO), with 3SR as the baseline and 4PR/4PR-X as the cases of interest. Our objective is to assess whether alternative emulators materially alter macroeconomic outcomes, ensuring a transparent foundation for comparison.

In what follows, we will demonstrate that emulator design, especially the addition of carbon reservoirs or time-dependent feedback, can significantly influence IAM projections. The following analyses provide an illustrative first look at richer climate modules and offer preliminary recommendations on when and how to incorporate additional reservoirs or dynamic operators into climate-economy models.

Our economic set-up consists of a single, infinitely lived, representative consumer and a single firm. We describe the equilibrium allocation as the solution to a social planner problem where the planner maximises a constant relative risk aversion (CRRA) utility function of per capita consumption, C_t/L_t . Here, C_t represents consumption, L_t denotes labour, with a constant intertemporal elasticity of substitution (IES), $\psi > 1$, and a discount factor, $0 < \beta < 1$.

Equations (19)–(23) below jointly define the social planner's optimization problem. The value of the lifetime utility, V_0 is given by the following expression:

$$V_0 = \max_{\{C_t, \mu_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \frac{\left(\frac{C_t}{L_t}\right)^{1-1/\psi} - 1}{1-1/\psi} L_t \quad (19)$$

$$\text{s.t.} \quad K_{t+1} = (1 - \Theta(\mu_t) - \Omega(\mathbf{T}_t^A)) K_t^\alpha (A_t L_t)^{1-\alpha} + (1 - \delta)K_t - C_t. \quad (20)$$

²³ Additional numerical experiments drawn from the climate-science literature, which validate the full emulator framework, that is, combining the carbon cycle and temperature modules discussed in Sections 2.1 and 2.2, are reported in Online Appendix A.

²⁴ All IAMs were solved using 'Deep Equilibrium Nets', a deep-learning method for dynamic stochastic models; see Azinovic *et al.* (2022) for the general approach and Friedl *et al.* (2023) and Kübler *et al.* (2025) for its IAM application.

Climate Equations (1), (15) and (16)

$$0 \leq K_{t+1} \quad (21)$$

$$0 \leq \mu_t \leq 1, \quad (22)$$

where emissions that enter (1) are defined by:

$$e_t = \sigma_t Y_t^{\text{Gross}}(A_t, K_t, L_t)(1 - \mu_t) + E_t^{\text{Land}}, \quad (23)$$

and where E_t^{Land} are exogenous land emissions. Output $Y_t^{\text{Gross}}(A_t, K_t, L_t)$ is produced using a Cobb–Douglas technology, with capital K_t , total factor productivity (TFP) A_t , and labour L_t , where α represents capital elasticity. The capital stock depreciates at rate δ . Mitigation efforts, denoted by rate μ_t , are costly and reduce output by a factor $\Theta(\mu_t)$. Additionally, higher temperatures decrease output through the damage function $\Omega(\mathbf{T}_t^A)$ where $\mathbf{T}_t^A$ denotes atmospheric temperature.²⁵

We first consider the BAU case, where the social planner does not invest in mitigation μ_t . In this scenario, the planner only chooses the investment sequence, setting the mitigation sequence to zero at all times. Next, we consider the optimal mitigation case, where the planner simultaneously chooses investment in both capital and mitigation. In the optimal case, we examine the social cost of carbon, defined as the marginal cost of atmospheric carbon in terms of the numeraire good. Following the literature (e.g., Traeger, 2014; Cai and Lontzek, 2019), we define the social cost of carbon (SCC) as the planner's marginal rate of substitution between atmospheric carbon concentration and the capital stock:

$$SCC_t = -\frac{\partial V_t / \partial \mathbf{m}_t^A}{\partial V_t / \partial K_t}. \quad (24)$$

The SCC, as defined in Equation (24), equals the optimal carbon tax when $\mu_t < 1$.

In Sections 3.2 and 3.3, we present benchmark solutions for the BAU and optimal mitigation cases, respectively; Section 3.4 evaluates emulator performance under a hypothetical carbon capture and storage technology. We also analyse sensitivity to damage functions and discount rates (Online Appendix D) and report present-day simulation results in Online Appendix B.

3.2. Business as Usual

We begin by examining the BAU trajectory across the carbon cycle emulators introduced above. In this and all subsequent figures, the 3SR emulator is shown as a solid line, the 4PR emulator as a dashed line, and the 4PR-X emulator as a dotted line.²⁶ The extreme calibrations, MESMO and CLIMBER2-LPJ, are each shown as solid lines for their 3SR variants and dashed lines for their 4PR variants. In Figure 5(a), we observe that industrial emissions are identical across all carbon cycle models. However, the carbon mass in the land biosphere (Figure 5(b)) and atmosphere (Figure 5(c)) differ between the 4PR and 4PR-X models. This variation arises because the dynamic 4PR-X model simulates a decreasing land biosphere capacity, which leads to increased

²⁵ For the sake of brevity, we do not explicitly state the functional forms of some of the model equations, as well as parameterisation, unless it is necessary for presenting the results. Online Appendix C contains all the relevant information about the complete model specification, detailed parameterisation of all equations, including exogenous variables, and the procedure for integrating the climate emulator up to present-day conditions.

²⁶ Initial land biosphere stocks differ slightly because each emulator is integrated forward to present-day conditions with its own numerical solver; see Online Appendix C for details.

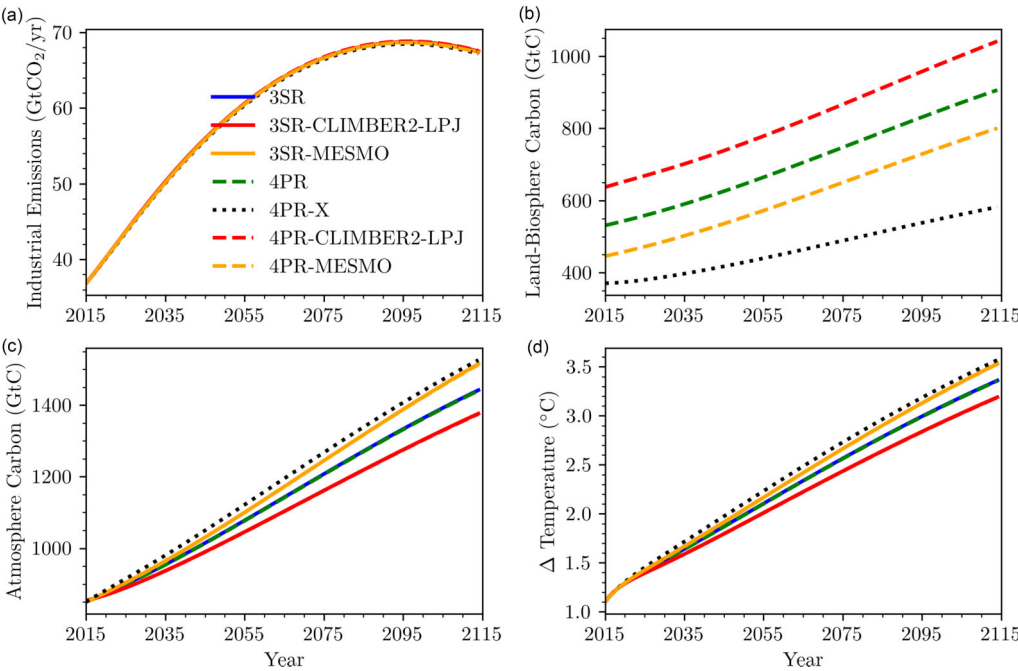


Fig. 5. The Simulation Results for the BAU Case ($\mu_t = 0$).

Table 3. Data for Atmospheric Carbon Mass ($\mathbf{m}_t^A$) and Atmospheric Temperature ($\mathbf{T}_t^A$) Across Selected Years.

Year	Variable	3SR (1)	4PR (2)	4PR-X (3)	diff (1) and (3)
2020	$\mathbf{m}_t^A_{2020}$ (GtC)	874.37	873.52	883.14	8.77 GtC (1.00%)
2050	$\mathbf{m}_t^A_{2050}$ (GtC)	1,046.07	1,045.01	1,085.54	39.47 GtC (3.77%)
2100	$\mathbf{m}_t^A_{2100}$ (GtC)	1,363.19	1,361.66	1,439.97	76.78 GtC (5.63%)
2020	$\mathbf{T}_t^A_{2020}$ (°C)	1.29	1.29	1.30	0.01 °C (0.99%)
2050	$\mathbf{T}_t^A_{2050}$ (°C)	1.99	1.98	2.10	0.11 °C (5.84%)
2100	$\mathbf{T}_t^A_{2100}$ (°C)	3.09	3.09	3.29	0.20 °C (6.34%)

Note: Absolute differences are 4PR-X relative to 3SR, with percentage differences in parentheses.

carbon uptake by other reservoirs, thereby raising atmospheric carbon mass and, consequently, temperature (Figure 5(d)). The rather low carbon content of the land reservoir of the 4PR-X model in the year 2015 (below 400 GtC and thus slightly outside the estimated range in Ciais *et al.*, 2014) is linked to the fact that we chose to simply remove all land use change emissions from the land biosphere equilibrium mass instead of applying some scaling as discussed in Section 1.1. Notably, the extreme variants 3SR-MESMO and 4PR-MESMO yield virtually the same atmospheric carbon burden and temperature rise as the 4PR-X emulator (Figure 5(c), (d)). This similarity highlights that a diminished land biosphere sink can produce climate outcomes comparable to the most pessimistic warming scenarios.

Table 3 presents the exact values for atmospheric carbon mass and temperatures in 2020, 2050 and 2100, along with the percentage differences between the 3SR and 4PR-X models.

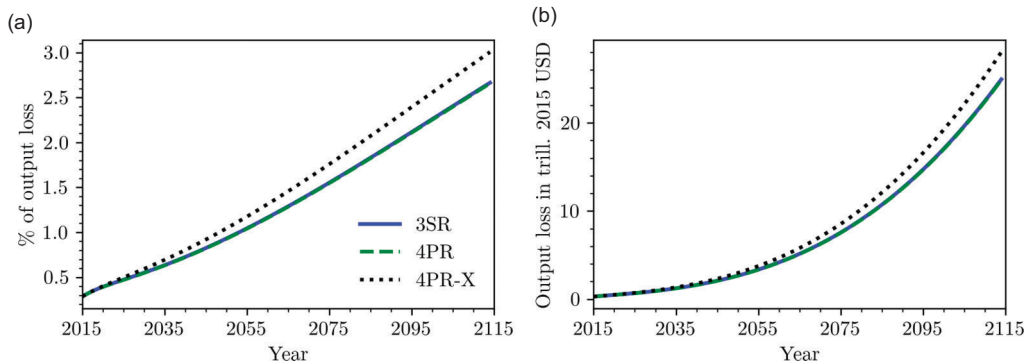


Fig. 6. Share of Output Lost to Temperature Damages (a) and the Resulting Monetary Loss (b).

In the BAU case, the carbon masses and temperatures in 2020 show no significant variation across the different carbon cycle models. However, by 2050, differences become substantial: the dynamic 4PR-X model shows nearly 4% higher atmospheric carbon and approximately 6% higher atmospheric temperature than the 3SR model.

In 2100, the dynamic scenario experiences an additional 0.2°C of warming relative to the static baseline (see Table 3). To translate this temperature increment into economic terms, we measure the fraction of gross output lost to climate damages with the function $\Omega(\mathbf{T}_t^A)$, parameterised by the coefficients $\psi_1 = 0$ and $\psi_2 = 0.00236$. Specifically,

$$\Omega(\mathbf{T}_t^A) = \psi_1 \mathbf{T}_t^A + \psi_2 (\mathbf{T}_t^A)^2, \quad (25)$$

so that higher atmospheric temperatures T_t^{AT} lead to an increasingly non-linear share of output loss (cf. Equation (25)). The associated absolute loss is obtained by multiplying this share with gross output,

$$D_t = \Omega(\mathbf{T}_t^A) K_t^\alpha (A_t L_t)^{1-\alpha}, \quad (26)$$

and is reported in trillion 2015 US dollar. Figure 6(a) contrasts the evolution of the damage share Ω with the corresponding monetary loss D_t , given by Equation (26), in Figure 6(b).

Figure 6 shows that both the damage share and the associated monetary loss track the temperature trajectory across all models; consequently, losses are largest in the dynamic 4PR-X case and smaller in the static 3SR and 4PR cases.

In the dynamic 4PR-X model, the additional 0.2°C of warming relative to the static emulators increases damages in 2050 by 11.7%, and in 2100 by 12.7% (see Table 4). This underscores that deforestation-induced warming imposes substantial additional economic damages.

3.3. Optimal Mitigation

Figure 7 traces the same variables under the optimal mitigation policy. The ordering of outcomes, 4PR-X above 4PR and 3SR, mirrors the BAU experiment. The optimal run additionally displays the mitigation rate and the SCC (Figure 7(f)). Both series are consistently higher in the dynamic 4PR-X emulator than in the static 3SR and 4PR counterparts, reflecting its warmer temperature

Table 4. *Data for Share of Output Loss and Equivalent Monetary Loss Across Selected Years.*

Year	Variable	3SR (1)	4PR (2)	4PR-X (3)	diff (1) and (3)
2020	$\Omega(T_t^A)$ p.p.	0.3938	0.3922	0.4016	0.0078 (1.98%)
2050	$\Omega(T_t^A)$ p.p.	0.9321	0.9288	1.0442	0.1121 (12.02%)
2100	$\Omega(T_t^A)$ p.p.	2.2603	2.2542	2.5562	0.2958 (13.08%)
2020	D_t (Trill 2015 US\$)	0.4881	0.4858	0.4965	0.0084 (1.73%)
2050	D_t (Trill 2015 US\$)	2.6684	2.6578	2.9802	0.3118 (11.68%)
2100	D_t (Trill 2015 US\$)	17.0611	17.0127	19.2261	2.165 (12.68%)

Note: Absolute differences are 4PR-X relative to 3SR, with percentage differences in parentheses.

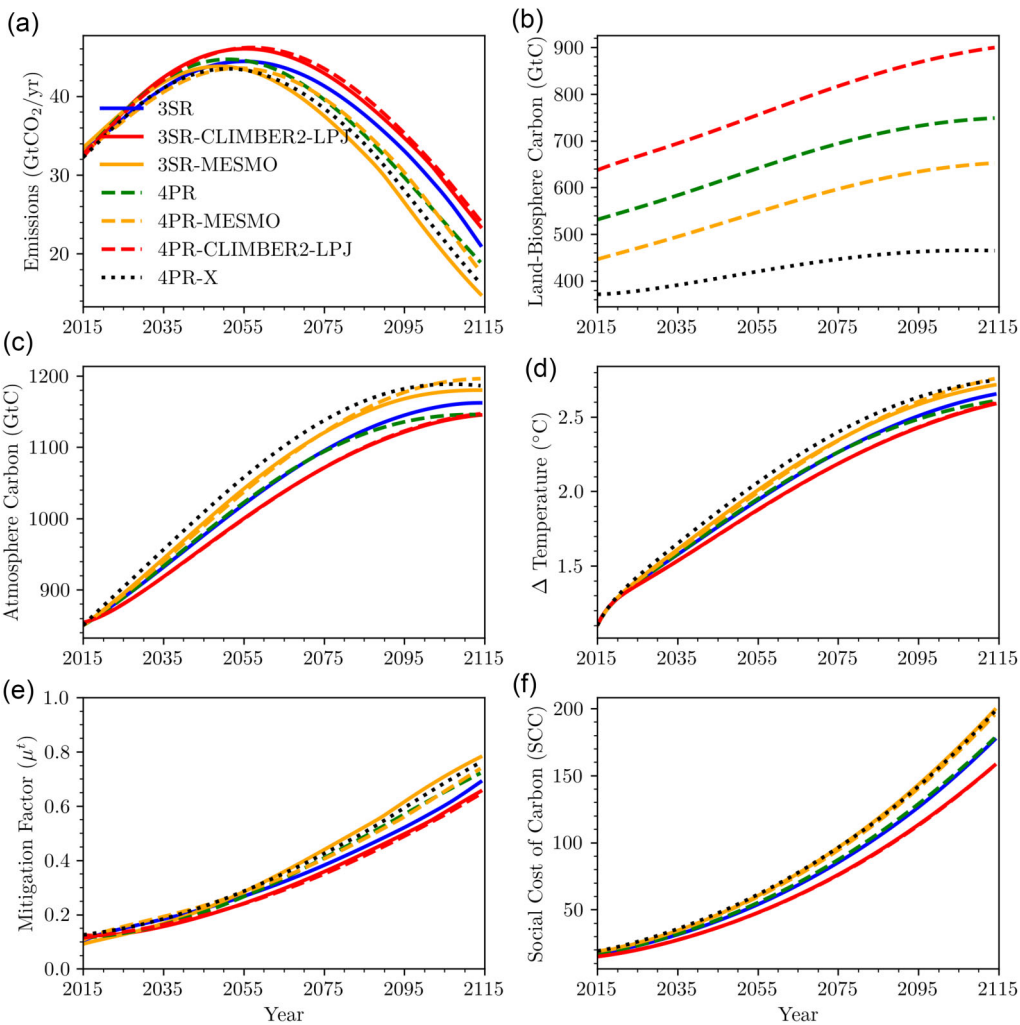


Fig. 7. *Simulation Results for the Optimal Abatement Scenario.*

Table 5. *Social Cost of Carbon (2015 US Dollar per Tonne of CO₂) Across Different Years.*

Year	Variable	3SR (1)	4PR (2)	4PR-X (3)	diff (1) and (3)
2020	SCC	19.85	19.72	22.20	2.35 (11.87%)
2050	SCC	47.47	48.57	54.07	6.60 (13.90%)
2100	SCC	138.58	140.82	155.45	16.87 (12.18%)

Note: Absolute differences are 4PR-X relative to 3SR, with percentage differences in parentheses.

Table 6. *Atmospheric Temperature Data (T_t^{AT}) Across Different Years.*

Year	Variable	3PR (1)	4PR (2)	4PR-X (3)	diff (1) and (3)
2020	T_{2020}^A (°C)	1.22	1.22	1.23	0.01 °C (1.11%)
2050	T_{2050}^A (°C)	1.14	1.14	1.22	0.08 °C (6.72%)
2100	T_{2100}^A (°C)	1.10	1.10	1.18	0.08 °C (7.09%)

Note: Absolute differences are 4PR-X relative to 3PR, with percentage differences in parentheses.

path. Table 5 quantifies these differences: the 4PR-X SCC exceeds the 3SR value by 11.9% in 2020, and by 13.9% in 2050. As in the BAU case, the extreme calibrations 3SR-MESMO and 4PR-MESMO closely track 4PR-X, confirming that a depleted land biosphere sink, e.g., extensive deforestation, can induce effects comparable to an extreme warming scenario.

Our benchmark comparison reveals no substantive difference between the static three- and four-reservoir emulators: adding a land biosphere reservoir has no impact on either BAU or optimal policy outcomes. By contrast, allowing the land biosphere stock in the four-reservoir model to decline over time raises BAU temperatures relative to the static cases. Explicitly modelling deforestation dynamics further increases the SCC, implying that land use change materially shifts the optimal mitigation strategy and requires a higher carbon price to curb damages.

The mechanism is straightforward. Deforestation diminishes the land biosphere sink’s uptake capacity, so carbon that would otherwise have been sequestered remains in the atmosphere, raising temperatures and amplifying damages. This feedback translates into a higher optimal carbon price; omitting land use change, therefore, leads to a systematic underpricing of carbon.

3.4. *Carbon Capture and Storage*

To illustrate how emulator design can significantly impact model outcomes, we examine a carbon capture and storage (CCS) scenario. CCS removes CO₂ directly from the atmosphere, thereby targeting the primary driver of warming. If deforestation is ignored, however, the apparent effectiveness of CCS can be overstated: deforestation weakens the land biosphere sink, leaving more carbon in the atmosphere, so CCS must compensate for emissions that would otherwise have been sequestered naturally.

We analyse this effect within the DICE-2016 framework by implementing a highly stylised policy that sets the mitigation rate to unity from the initial period, representing full deployment of the backstop technology and eliminating industrial CO₂ emissions. Figure 8 plots the resulting atmospheric carbon and temperature trajectories for each emulator, and Table 6 lists the corresponding temperature values.

Consistent with the BAU and optimal policy experiments, adding a static land biosphere reservoir leaves climate trajectories practically unchanged. By contrast, when the four-reservoir emulator incorporates a declining equilibrium land sink, it exhibits the same qualitative diver-

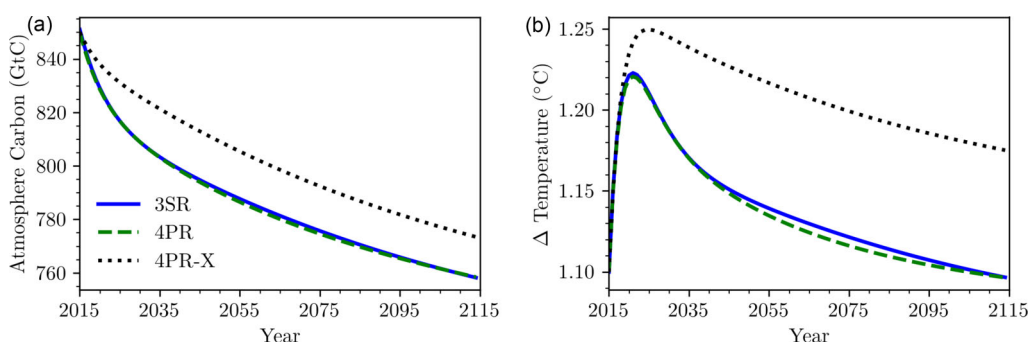


Fig. 8. Atmospheric Carbon Concentrations (a) and Atmospheric Temperature (b) Under the Full Abatement Scenario ($\mu_t = 1$).

gence observed earlier: even with full carbon capture in place, continued deforestation raises atmospheric temperatures by roughly 7% relative to the static cases by 2050.

Under the full mitigation, the 4PR-X model projects higher atmospheric carbon and temperature than 3SR, due to its reduced biospheric uptake.²⁷ This underscores that even with ambitious CCS policies, failing to model biosphere dynamics may lead to overly optimistic projections of climate outcomes and underestimation of required mitigation levels. Our findings indicate that CCS can succeed only if natural carbon sinks are preserved. Land degradation and deforestation erode the land-biosphere's sequestration capacity, undermining the net benefit of engineered carbon removal. Policies that protect or enhance terrestrial sinks are therefore an essential complement to large-scale CCS deployment.

4. Linking Global to Local Temperatures and Damages

Economic models that resolve climate impacts across space require equally resolved projections of key climatic drivers, in particular regional temperatures (in $^{\circ}\text{C}$) and their future changes, to quantify local damages and related outcomes (e.g., Krusell and Smith, 2022; Cruz and Rossi-Hansberg, 2024; Desmet and Rossi-Hansberg, 2024; Kotlikoff *et al.*, 2024). In this section, we present a computationally efficient procedure, grounded in state-of-the-art climate science, for deriving regional temperatures and temperature changes from global climate emulators (see Section 2).

A widely used and inexpensive way to obtain such projections is *pattern scaling*, a form of statistical downscaling that expands a change in global mean temperature into a gridded warming pattern (see Tebaldi and Arblaster, 2014; Kravitz *et al.*, 2017; Lynch *et al.*, 2017; Mathison *et al.*, 2024; 2025, and references therein). The resulting pattern can subsequently be aggregated to any geographical units the spatially resolved models require.

²⁷ In the 4PR-X model, carbon emissions from land use change are assumed to reduce the carbon storage capacity of the land biosphere concurrently and by an equal amount. The underlying reasoning is that, for example, deforestation, not only results in carbon emissions, but it also means the forest will no longer be there as a carbon reservoir. The shrinking carbon-holding capacity of the land biosphere implies that a higher share of carbon emissions will be directed to the remaining reservoirs, notably the atmosphere. This effect of land use change accumulates over time. Consequently, the difference between the 3SR and 4PR-X model, which are both calibrated against PI conditions, increases with time.

Because both the global mean temperature trajectory and the warming pattern pertain to an as-yet unobserved future, they must be taken from earth system model (ESM) simulations. The pattern is therefore not unique; it inherits the spread of the underlying ESM ensemble, just as an emulator for global mean temperature inherits the calibration choice of its driving ESM(s) (see Section 3). Selecting a particular pattern thus introduces an additional source of uncertainty driven by model choice. When absolute temperatures (rather than anomalies) are needed, the pattern must be anchored to present-day observations, ensuring consistency with the empirical climate record.

In what follows, Section 4.1 formalises the pattern-scaling method and states the governing equations. Section 4.2 quantifies (i) the spread of temperature change patterns across an ensemble of ESMs and (ii) the region-specific absolute temperatures projected for 2100. Building on these results, it also outlines a procedure for deriving region-specific temperatures that are consistent with both empirical observations and the ESMs. Section 4.3 links these local temperature projections to illustrative local impacts and damages in the year 2100.

4.1. Basics of Pattern Scaling

Pattern scaling, introduced by Santer *et al.* (1990), establishes a relationship between global and local temperature changes using large-scale ESM outputs. Specifically, it relates the change in global mean near-surface air temperature, ΔT^{AT} , to the local mean near-surface air temperature change, ΔT^z , at a specific location z (representing areas potentially as fine as $1^\circ \times 1^\circ$). Temperature changes are typically computed after temporal smoothing, often by averaging over 10 or 20 years, to remove high-frequency interannual variability. A linear relationship is generally considered an adequate approximation for temperature, as well as for other variables such as precipitation (Lynch *et al.*, 2017; Pfahl *et al.*, 2017; Mathison *et al.*, 2025; Munday *et al.*, 2025) and sea level change (Bilbao *et al.*, 2015). Limitations of the linearity assumption can arise in the presence of highly localised radiative forcing agents, such as anthropogenic aerosols, or when multi-century timescales are considered (see, e.g., Rugenstein *et al.*, 2016); however, these issues are typically irrelevant for many economic applications.

Formally, the linear pattern-scaling approach used in this article expresses the local temperature change ΔT^z , measured in $^\circ\text{C}$, at location z as:

$$\Delta T^z = \Delta T^{\text{AT}} \beta^z, \quad (27)$$

where β^z denotes the spatially resolved temperature change pattern. Because β^z relates to future climate change, it cannot be derived from observations, but must be obtained from model simulations, notably ESMs, thereby inheriting ESM-associated uncertainties.

For the subsequent analysis, this work relies on a publicly accessible library of temperature change patterns β^z by Lynch *et al.* (2017).²⁸ They applied least squares regressions to CMIP5 data from future RCP8.5 scenarios across forty-one different ESMs. For each model, a least squares regression relates the time series of annual global mean temperature change ΔT_t^{AT} to the time series of the location-specific (z), two-dimensional gridded latitude-longitude temperature change ΔT_t^z via:

$$\Delta T_t^z = \alpha^z + \beta^z \cdot \Delta T_t^{\text{AT}} + \epsilon_t^z. \quad (28)$$

²⁸ https://github.com/JGCRI/CMIP5_patterns.

In this regression, the term β^z represents the desired temperature change pattern. It is a two-dimensional field of regression slopes quantifying the temperature change at position z relative to the global mean temperature change; thus, it has physical dimensions of degrees Celsius (local change) per °C (global change). The other terms in (28), the y-intercept α^z (which Lynch *et al.*, 2017 assume to be zero) and the residual ϵ_t^z , are not directly used in the pattern scaling application itself. The forty-one scaling patterns derived from the 41 ESMs are qualitatively similar, but quantitatively distinct, reflecting inter-model differences—we provide concrete examples of this below. As there is often no clear basis for selecting one ESM pattern as superior, real-world applications should consider employing multiple patterns to assess the robustness of results against this pattern uncertainty.

Local damage functions, however, often require projections of future *absolute temperature* patterns, T_{abs}^z (e.g., Krusell and Smith, 2022; Desmet and Rossi-Hansberg, 2024, and references therein). Absolute temperatures are typically more challenging to quantify accurately than temperature changes, both in observations and models. ESMs, for instance, are primarily designed and evaluated based on their ability to simulate temperature changes relative to a baseline (e.g., pre-industrial) rather than absolute temperatures (e.g., Mauritsen *et al.*, 2012).

Obtaining projections of absolute temperatures using pattern scaling requires anchoring the *temperature change* pattern β^z to an *absolute temperature pattern* $T_{\text{abs},c}^z$ from a historical reference period (typically a 30-year climatological average). The future absolute temperature at time t is then calculated as:

$$T_{\text{abs},t}^z = T_{\text{abs},c}^z + \Delta T_t^{\text{AT}} \cdot \beta^z, \quad (29)$$

where ΔT_t^{AT} in Equation (29) represents the global mean temperature change between the chosen historical reference period and the future time of interest, t .

There is no single best choice for the historical anchor pattern $T_{\text{abs},c}^z$. Since this pattern pertains to the historical period, options include observation-based datasets or model-based climatologies. For economic applications where proximity to real-world conditions is often crucial, using an observation-based historical absolute temperature dataset for $T_{\text{abs},c}^z$ is a common and justifiable choice. Several publicly available gridded datasets of observed absolute surface air temperature exist.²⁹ Among the highest-regarded datasets currently available is the ERA5 re-analysis (Hersbach *et al.*, 2020). This dataset is used below to derive the gridded 1991–2020 climatological mean absolute surface air temperature, $T_{\text{abs},c}^z(\text{ERA5})$, shown in Figure 9(a). The global mean value of $T_{\text{abs},c}^z(\text{ERA5})$ over this period is 14.4°C, consistent with the global mean temperature estimate for 1991–2020 published by the European Union’s Copernicus Climate Change Service.³⁰ Thus, in practical applications, local absolute temperature can be computed as

$$T_{\text{abs},t}^z = T_{\text{abs},c}^z(\text{ERA5}) + \Delta T_t^{\text{AT}} \beta^z, \quad (30)$$

where ΔT_t^{AT} denotes the global mean temperature change provided by the climate emulator (see Section 2 and Table 3).

An alternative approach for choosing $T_{\text{abs},c}^z$, described by Lynch *et al.* (2017), prioritises *internal consistency* within each ESM and anchors both, warming pattern and historical baseline, to

²⁹ Among them are re-analysis data from ERA5 (<https://www.ecmwf.int/en/forecasts/dataset/ecmwf-reanalysis-v5>), JRA-3Q (https://jra.kishou.go.jp/JRA-3Q/index_en.html), MERRA-2 (<https://gmao.gsfc.nasa.gov/reanalysis/merra-2>), and NCEP (<https://psl.noaa.gov/data/gridded/data.ncep.reanalysis2.html>) or for land only temperatures the CRU data (<https://crudata.uea.ac.uk/cru/data/hrg>).

³⁰ <https://climate.copernicus.eu>.

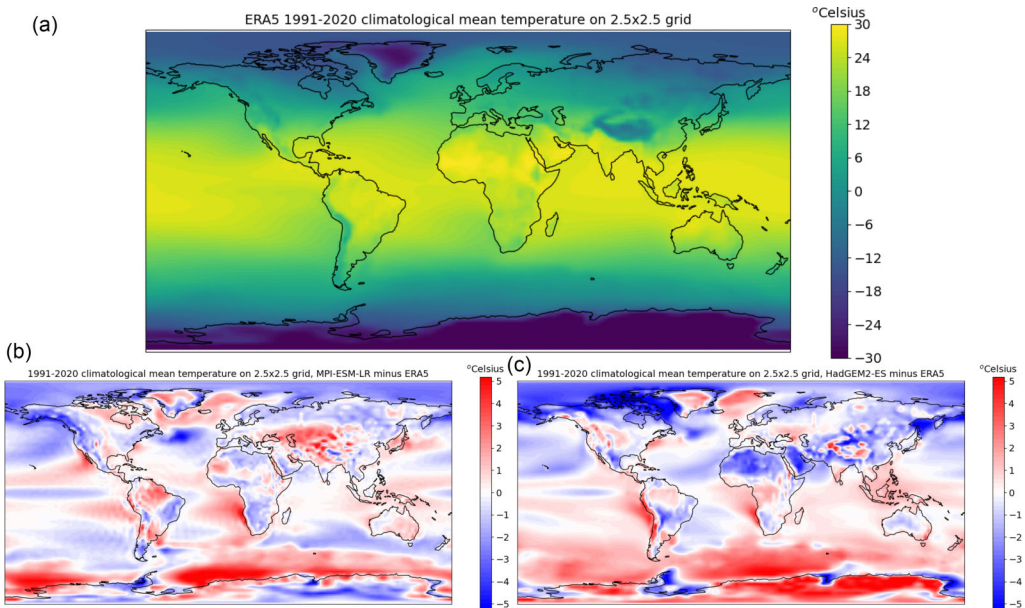


Fig. 9. Panel (a): 30-Year Climatological Mean Near-Surface Air Temperature (1991–2020) from the ERA5 Re-Analysis, Taken as the Observational Reference. Panels (b) and (c): temperature biases of two CMIP5 earth system models with respect to that reference, shown as model climatology minus ERA5. (b) MPI-ESM-LR (Max-Planck-Institute earth system model, low-resolution). (c) HadGEM2-ES (Hadley Centre Global Environmental Model, version 2–earth system). Positive values indicate regions where the ESMs are warmer than ERA5.

output from the *same* ESM run. The ESM specific historical 30-year averaged (1961–90) absolute temperature climatologies, $T_{\text{abs},c}^z(\text{ESM})$, exhibit global mean values ranging from 12.5°C to 15.3°C , compared with the observed global mean estimate for 1961–90 of $14.0 \pm 0.5^\circ\text{C}$ (Jones *et al.*, 1999). The offset between observations and a given ESM can be removed by rescaling: $T_{\text{abs},c}^z(\text{ESM}) + \Delta T_{\text{obs,ESM}} \beta^z(\text{ESM})$, where $\Delta T_{\text{obs,ESM}} = 14.0 - T_{\text{abs},c}^z(\text{ESM})$. The resulting temperature fields are specific to the ESM's climate and generally differ from fields anchored directly to observations. Figure 9 illustrate the differences between the 1991–2020 climatologies based on ERA5 (Figure 9(a)) and the appropriately rescaled climatologies based on two example ESMs (Figure 9(a), (b)), namely MPI-ESM-LR (Max-Planck-Institute earth system model, low-resolution) and HadGEM2-ES (Hadley Centre Global Environmental Model, version 2–earth system). Both models are well established in climate science and display somewhat different warming patterns. In conclusion, various options exist for selecting $T_{\text{abs},c}^z$, each potentially suitable for different purposes; the choice should be made carefully based on the specific requirements of the application.

Combining pattern scaling, as described, with an emulator for global mean temperature change, such as CDICE (Folini *et al.*, 2025), provides a computationally efficient and transparent method for generating spatially resolved temperature fields suitable for input into damage functions. As detailed above, these temperature fields are subject to uncertainties stemming from the choice and calibration of the global temperature emulator, the selection of the scaling pattern β^z , and the choice of the anchoring pattern $T_{\text{abs},c}^z$. A key advantage of this component-based approach is

that the influence of each source of uncertainty can be explored separately and transparently, as demonstrated below. While other publicly accessible tools for pattern scaling exist (e.g., those presented by Beusch *et al.*, 2020; Hernanz *et al.*, 2023), they may be less readily suited for such component-wise sensitivity analysis due to differences in their structure or complexity.³¹

A technical summary of how to do pattern scaling reads as follows.

- *Select a warming pattern.* Choose β^z from one of the available earth system models.
- *Select an absolute temperature pattern for anchoring.* When absolute future temperature fields $T_{\text{abs},t}$ are required, choose $T_{\text{abs},c}^z$ from among observation-based datasets, for example ERA5 re-analysis.
- *Select a baseline period.* Decide on the baseline period relative to which one global mean temperature change can be computed (employing your CE) and over which you average the absolute temperature fields needed for pattern scaling, for example, the 30-year period 1991–2020.
- *Compute future (absolute) temperatures.* Combine the chosen β^z and $T_{\text{abs},c}^z$ with the projected global mean warming ΔT_t^{AT} to obtain

$$T_{\text{abs},t}^z = T_{\text{abs},c}^z + \Delta T_t^{\text{AT}} \beta^z.$$

4.2. Spatial Temperature Change Patterns

This section illustrates the uncertainty associated with the parameter β^z —see (27)—by demonstrating how a one-degree Celsius change in global mean temperature translates into region-specific warming, and by assessing the sensitivity of these estimates to the underlying earth system models (ESMs). Furthermore, the section addresses the implications of selecting $T_{\text{abs},c}^z$, which is necessary to generate absolute temperature patterns. For illustration, we use two models from the forty-one ESMs presented in Lynch *et al.* (2017), namely MPI-ESM-LR and HadGEM2-ES (see Figure 9).

To ensure tractability, regional analyses often aggregate smaller areas (e.g., $1^\circ \times 1^\circ$ grid cells) into larger units. In this study, we employ the WGI (v4) reference regions (Iturbide *et al.*, 2020). Defined on geographical and climatological criteria, these regions are widely used in recent IPCC reports and climate-modelling studies, yet they remain relatively uncommon in the economics literature.³² Our analysis focuses exclusively on the land areas of these regions, as shown in Figure 10. The regions are listed and explained in Table 12 of Online Appendix E. The regional partition adopted here balances spatial resolution against computational tractability. Nevertheless, the optimal degree of regional aggregation ultimately depends on the specific objectives of the study at hand.

Figure 11 shows the geographical warming patterns (β^z) per degree of global mean temperature increase for the MPI-ESM-LR and HadGEM2-ES climate models. Several key features are apparent: warming is particularly strong across Arctic regions and most land areas experience warming rates exceeding the global average ($\beta^z > 1$). Significant spatial heterogeneity exists, leading to markedly different warming rates even in geographically neighbouring regions, such

³¹ See <https://github.com/ahernanz/pyClim-SDM> and <https://github.com/MESMER-group/mesmer-openscmrunner>, respectively.

³² See https://github.com/IPCC-WGI/Atlas/blob/main/reference-regions/IPCC-WGI-reference-regions-v4_coordinates.csv for details.

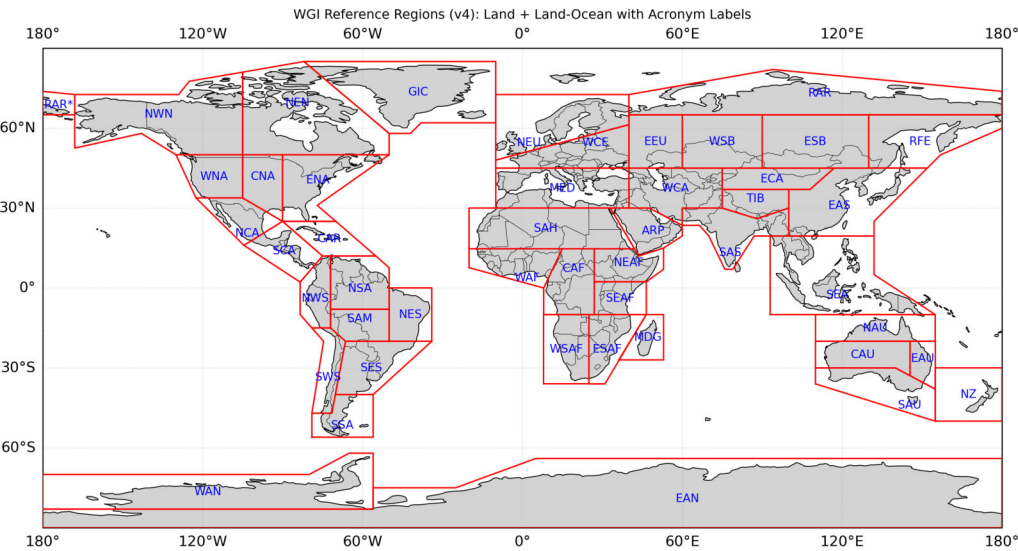


Fig. 10. WGI (v4) Reference Land and Land-Ocean Regions (Iturbide et al., 2020).

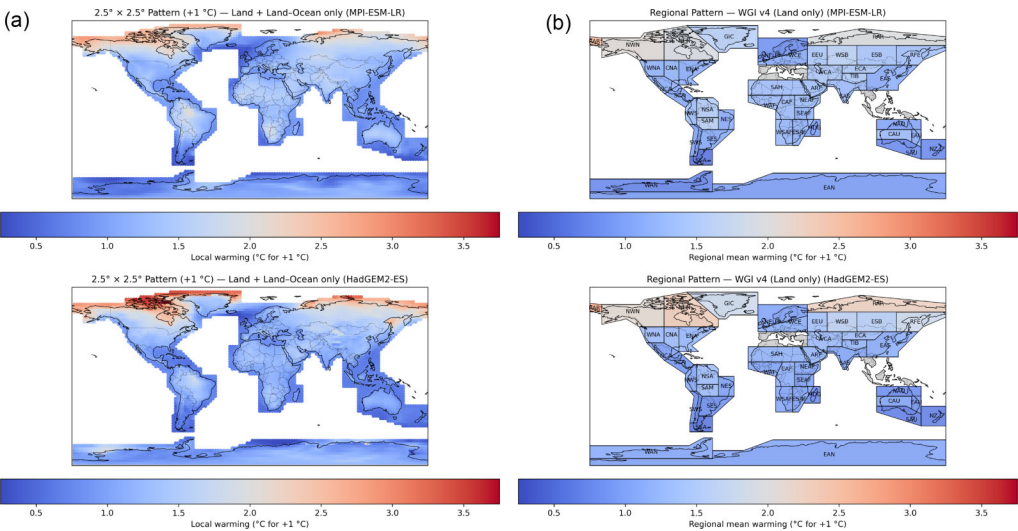


Fig. 11. Spatial Warming Patterns β^z Corresponding to a 1°C Increase in Global Mean Temperature, Shown as Gridded Maps (a) and Regionally Aggregated Maps (b). The patterns are generated using pattern scaling based on the ESMs MPI-ESM-LR (top row) and HadGEM2-ES (bottom row).

as the contrast observed within South America between the South America monsoon (SAM) and northeastern South America (NES) regions. Furthermore, the warming patterns differ notably between the two models. For instance, MPI-ESM-LR projects stronger warming than HadGEM2-ES in areas like western Brazil and parts of South Africa, while HadGEM2-ES shows greater warming in northern Canada. These inter-model discrepancies persist when aggregated to the reference region level. Compared to HadGEM2-ES, the MPI-ESM-LR model shows greater

Table 7. *Subset of IPCC WGI Version-4 Land-Regions Used in this Study.*

Acronym	RefName	β				$T_{\text{abs}}(2100)$			
		MPI	Had	Min	Max	MPI+M	Had+H	MPI+E	Had+E
CNA	C. North America	1.21	1.35	0.89	1.84	14.50	14.18	14.51	14.71
ECA	E. Central Asia	1.46	1.38	1.09	1.61	10.93	7.11	8.49	8.37
ARP	Arabian Peninsula	1.36	1.25	1.00	1.50	29.24	26.52	28.80	28.63
EEU	E. Europe	1.42	1.37	1.06	1.98	8.84	7.69	7.03	6.93
SAS	South Asia	1.30	1.07	0.86	1.33	28.02	25.68	27.63	27.25

Notes: Columns 1–2 list the official acronym and reference name. Columns 3–4 give the regional warming per degree of global mean warming, β^z , estimated by MPI-ESM-LR (‘MPI’) and HadGEM2-ES (‘Had’). Columns 5–6 show the minimum and maximum β^z values across the full 41-member CMIP5 ensemble, thus indicating the structural spread among models. Columns 7–10 report the projected near-surface air temperature in the year 2100, $T_{\text{abs}}(2100)$, assuming a global mean warming since pre-industrial times of $\Delta T^{\text{AT}} = 2.65^\circ\text{C}$ (scenario 4PR-X; see Figure 7). Values labelled ‘MPI+M’ and ‘Had+H’ use each model’s own 1961–90 climatology, $T_{\text{abs},c}^z$ (ESM), as the baseline; values labelled ‘MPI+E’ and ‘Had+E’ employ the ERA5 1991–2020 climatology, $T_{\text{abs},c}^z$ (ERA5). All temperatures are expressed in degrees Celsius.

warming in northern South America (NSA), the South America monsoon (SAM) region, and western Southern Africa (WSAF), whereas northern Europe (NEN) warms more in HadGEM2-ES (by approximately 0.2 to 0.4 °C per degree of global warming). These differences may seem small, but they scale with global mean temperature change, and thus imply substantial cumulative impacts over time.

To illustrate this in more detail, Table 7 presents a subset of numerical values for five economically relevant WGI regions. The regions were selected for two reasons: (i) they display a strong warming amplification ($\beta^z \geq 1.30$ in either MPI-ESM-LR or HadGEM2-ES); and (ii) they coincide with economic blocs whose combined 2024 gross domestic product (GDP) exceeds US\$2 trillion.

Even within this limited sample, the regional warming factors vary appreciably: for the two illustrative ESMs the range is 1.21–1.46 (MPI-ESM-LR) and 1.07–1.38 (HadGEM2-ES), implying that these economies are projected to warm roughly 10–45% faster than the globe as a whole. The full forty-one-model ensemble widens the spread to 0.86–1.98, underscoring that the model choice for the warming pattern β^z adds a further uncertainty of up to $\sim 1^\circ\text{C}$ per degree of global warming (see Online Appendix F, Figure 32, and <https://github.com/JGCRI/CMIP5.patterns/>).

Turning to absolute temperature projections for 2100, given in the right-hand columns of Table 7, we find that the choice of the anchoring baseline (model-internal climatology versus ERA5) can shift regional means by up to about 2.5–3 °C. For example, east-central Asia warms to 10.9 °C when anchored to the MPI-ESM-LR climatology, but to only 8.5 °C when the same pattern is anchored to ERA5, a difference of 2.4 °C. Regional temperatures in the two columns using the same $T_{\text{abs},c}^z$ (ERA5) agree much better than those in the columns based on $T_{\text{abs},c}^z$ (ESM). Although the table is not exhaustive, the five regions serve as representative testbeds for sensitivity analyses that combine strong climate signals with large economic stakes. More details on β^z and its dependence on different regions and ESMs are provided in Online Appendix F, Figure 32.

In summary, the above data show that typical values for regionally averaged warming patterns β^z are in a range of $1.5 \pm 0.5^\circ\text{C per }^\circ\text{C}$ of global mean warming. Extreme values outside a range of 0.5–2.5 °C per °C do exist (see Online Appendix F, Figure 32). When realistic absolute future temperature patterns are required, the choice of $T_{\text{abs},c}^z$ is crucial, and we advocate the use of

Table 8. *Parameter Values for the Function $\tilde{D}(T_t^z)$.*

Parameter	Description	Value
d	Lower bound of $\tilde{D}(T_t^z)$ as T_t^z diverges from T^*	0.02
T^*	Optimal regional temperature where $\tilde{D}(T_t^z)$ is maximised	11.58
κ^+	Rate of decline of $\tilde{D}(T_t^z)$ for $T_t^z \geq T^*$	0.00311
κ^-	Rate of decline of $\tilde{D}(T_t^z)$ for $T_t^z < T^*$	0.00456

observation-based climatologies for this purpose. For large global mean temperature changes, differences in β^z will eventually dominate, as they scale with the magnitude of the change.

4.3. *Regional Damages from Global Warming by 2100*

Having discussed how global mean temperature trajectories translate into local temperatures, we now examine how these varying local climate outcomes lead to heterogeneous economic impacts across regions. The specification of spatially resolved damage functions is central to various policy questions. A common approach is to model damages as a function of local warming (see Section 4.2). Following Krusell and Smith (2022), many models assume an inverted U-shaped relationship between a region’s absolute temperature and productivity, calibrated so that the spatially resolved damage model replicates aggregate global damage estimates.³³ This approach aligns with evidence from climate science that relates current population density, crop production, and GDP to absolute temperature (e.g., Xu *et al.*, 2020).

In what follows, we adopt the formalism of Krusell and Smith (2022) to map the outputs of our global climate emulators and pattern-scaling procedure into region-specific damage estimates.³⁴ Within this framework, the climate component of regional total factor productivity (TFP), $\tilde{D}(T_t^z)$, is assumed to follow

$$\tilde{D}(T_t^z) = \begin{cases} (1 - d) \exp(-\kappa^+(T_t^z - T^*)^2) + d & \text{if } T_t^z \geq T^*, \\ (1 - d) \exp(-\kappa^-(T_t^z - T^*)^2) + d & \text{if } T_t^z < T^*, \end{cases} \tag{31}$$

where the parameter values for Equation (31) are reported in Table 8. Regional damages attributable to global warming are then calculated as the relative change of the climate component of TFP,

$$D(T_t^z) = \frac{\tilde{D}(T_t^z)}{\tilde{D}(T_{\text{Baseline}}^z)} - 1, \tag{32}$$

³³ This specification, also employed by Hassler *et al.* (2023) and Kotlikoff *et al.* (2024), can be parameterised to match aggregate damage patterns identified in the literature (Desmet and Rossi-Hansberg, 2024).

³⁴ As discussed in Section 4.2, absolute future temperatures are harder to quantify than temperature anomalies. We therefore apply the ‘ERA5-correction’ introduced in (30).

Table 9. *Temperature (in °C) ‘Today’, ERA5 1991–2020 Mean, and as Projected for 2100 in Selected South American Cities.*

City	Country	ERA5 [°C]	HadGEM2-ES	MPI-ESM-LR
			2100 [°C]	2100 [°C]
São Paulo	Brazil	21.61	23.44	23.57
Buenos Aires	Argentina	16.61	17.71	17.95
Rio de Janeiro	Brazil	22.15	23.43	23.60
Lima	Peru	14.25	15.59	16.30
Bogotá	Colombia	20.40	21.93	22.35
Santiago	Chile	9.74	11.18	11.58
Caracas	Venezuela	26.94	28.65	29.04
Quito	Ecuador	21.86	23.46	23.92
La Paz	Bolivia	11.58	13.37	13.89
Brasília	Brazil	24.38	26.30	26.25
Medellín	Colombia	20.40	21.93	22.35
Guayaquil	Ecuador	19.76	21.19	21.45
Asunción	Paraguay	22.83	24.48	25.32
Montevideo	Uruguay	16.31	17.20	17.59
Curitiba	Brazil	19.54	21.05	21.42

where T_{Baseline}^z denotes the local absolute temperature in the baseline period.³⁵ Values $D(T_t^z) > 0$ indicate an increase in TFP as the local temperature changes from T_{Baseline}^z to T_t^z , values $D(T_t^z) < 0$ indicate a decrease in TFP.

As this section is illustrative rather than exhaustive, we restrict our attention to South America. Specifically, we analyse two alternative regional temperature fields for the year 2100. Both fields are anchored to the ERA5 absolute temperature climatology for 1991–2020, denoted T_{Baseline}^z , and use warming patterns β^z from either HadGEM2-ES or MPI-ESM-LR. Table 9 lists present-day ERA5 temperatures together with 2100 projections for fifteen of South America’s largest cities. Corresponding damages (climate induced relative changes in TFP) are illustrated in Figure 12.

The city data reveal a wide present-day temperature span, from 9.74°C in Santiago to 26.94°C in Caracas. Because Santiago’s current climate lies below the optimal temperature of 11.58°C—see (31), moderate warming could raise local productivity. Such statements hinge on accurate present-day temperatures, underscoring the need to anchor pattern scaling in observation-based data such as ERA5. Between now and 2100, the projected warming ranges from roughly 1°C to 2°C. Model choice matters: MPI-ESM-LR yields larger increases for several cities, for example, Asunción warms by 0.84°C, about 50% more than under HadGEM2-ES, whereas others, such as Brasília, show similar warming between models.

Figure 12 shows projected 2100 regional damages under our two ESMs. The figure reveals that most of South America experiences net welfare losses, whereas segments of the Andes and the far south enjoy modest gains. Importantly, the projected damages are sensitive to model uncertainty: the MPI-ESM-LR map shows larger losses than its HadGEM2-ES counterpart. The spatial distribution differs: for instance, the areas of greatest loss are centred on the Amazon under HadGEM2-ES, but extend into Paraguay and Colombia under MPI-ESM-LR (see Table 9). Such

³⁵ Many studies (see, e.g., Krusell and Smith, 2022; Kotlikoff *et al.*, 2024, and references therein) use Nordhaus’ GEcon database or similar sources to construct GDP-weighted regional temperature patterns (<https://gecon.yale.edu>, GEcon 4.0 for 2005). GDP weighting ensures that grid cells with greater economic activity exert a larger influence on the regional mean. For example, Canada’s average is drawn towards its populous southern corridor, whereas an unweighted mean would be dominated by sparsely populated Arctic cells. A detailed comparison of weighting schemes is beyond the scope of this article.

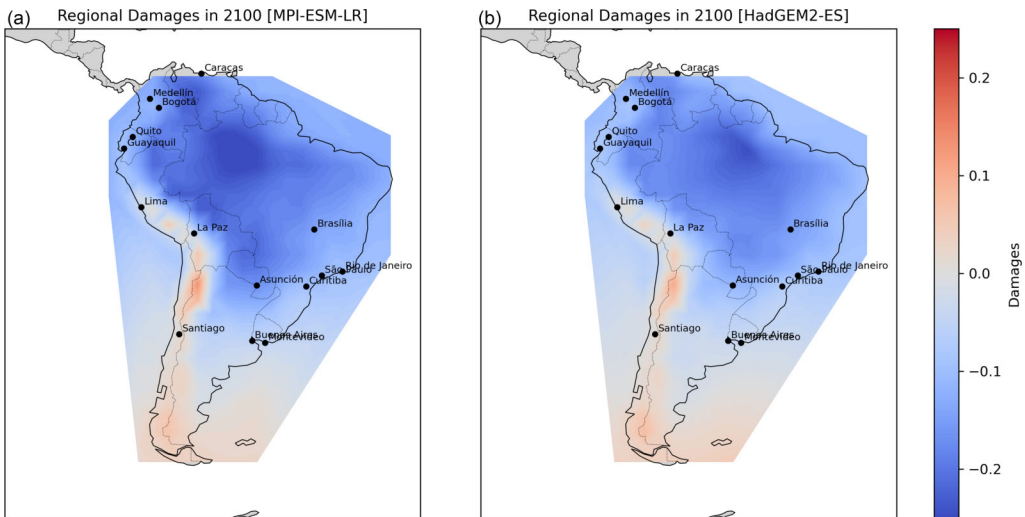


Fig. 12. Damages, Expressed as $\tilde{D}(T_{2100}^z)/\tilde{D}(T_{Baseline}^z) - 1$, Assuming a Global Mean Warming of 2.65°C Since Pre-Industrial Times ($\approx 1.55^\circ\text{C}$ since 2015) Under the 4PR-X Scenario. Absolute temperatures stem from ERA5 ($T_{abs,c}^z$); pattern factors β^z are taken from MPI-ESM-LR (a) or HadGEM2-ES (b). Lighter (darker) shading indicates an increase (decrease) in TFP due to global warming by 2100.

discrepancies highlight the need for caution when interpreting spatially resolved yet regionally averaged results.

When the damage function (32) is used to classify relative winners and losers under global warming, the two factors on the right-hand side of the pattern scaling in (27) interact in a non-trivial manner. Figure 13 illustrates this interaction. The absolute present-day temperature (shown on the x -axis), introduced via $T_{abs,c}^z$, determines whether a location can benefit from an increase in the global mean temperature: gains are possible only if $T_{abs,c}^z < 11.58^\circ\text{C}$. It also determines the impact of any given future temperature change. For example, a warming of 2.5°C results in different relative changes of damages, depending on present-day conditions, having a smaller effect when today's temperature is around 15°C than when the present-day temperature is 25°C . Put differently: how uncertainties in future warming (β^z) translate into relative changes in damages also depends on present-day temperatures ($T_{abs,c}^z$). While $T_{abs,c}^z$ can be measured reliably from observation-based products such as ERA5, uncertainty in the warming pattern, derived from the emulator-based global mean change multiplied by β^z , is unavoidable because it depends on the chosen ESM.

Therefore, these two components of pattern scaling play different roles. Observation-based absolute temperatures $T_{abs,c}^z$ (ERA5) anchor the analysis in the present-day climate and establish the potential for benefits, whereas the model-dependent warming patterns β^z drive the spread of outcomes. The transparency of the pattern-scaling framework makes this distinction explicit.

5. Conclusion

This paper presents an adaptable computational framework that enables IAM practitioners to construct transparent, computationally efficient climate box emulators, with up to four reservoirs,

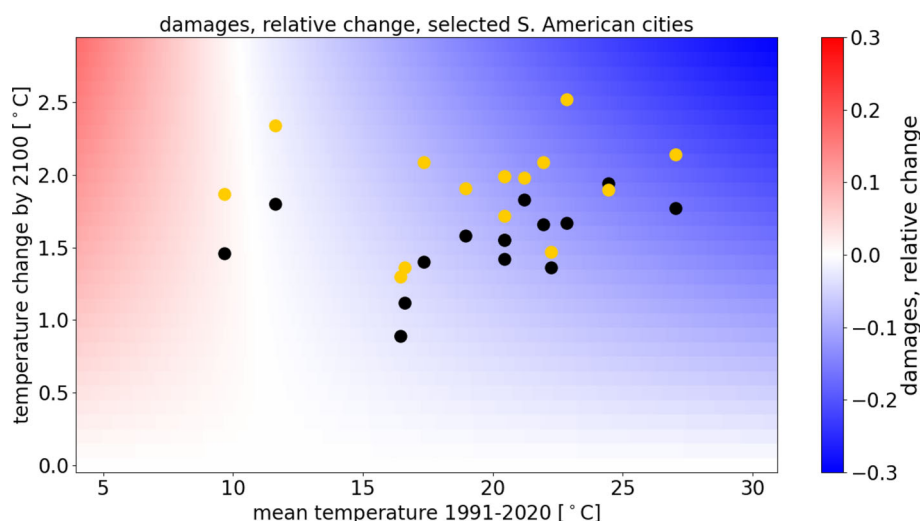


Fig. 13. Relative Change in Damages $\tilde{D}(T_{2100}^z)/\tilde{D}(T_{Baseline}^z) - 1$ as a Function of Present-Day (x-axis) And Future (y-axis) Temperatures, Based on (32). Coloured points correspond to the cities listed in Table 9, with HadGEM2-ES (dark markers) and MPI-ESM-LR (light markers) projections for 2100. Lighter (darker) background shading indicates an increase (decrease) in TFP.

tailored to their research questions. By focusing on linear box model structures, the framework supports rapid calibration while preserving interpretability, physical consistency, and seamless integration into economic and policy analyses.

To illustrate how calibration targets, emulator configuration, and hyperparameter selection affect economic outcomes, we compare three carbon cycle emulators calibrated to pre-industrial and present-day conditions. Our results show that incorporating a dynamically evolving land biosphere reservoir materially alters climate-economic projections. The PI-calibrated 4PR model, which treats the land reservoir as static, behaves similarly to the simpler 3SR configuration, whereas the 4PR-X variant, which endogenises land use change, yields markedly higher atmospheric CO₂ concentrations and temperatures under both RCP and business-as-usual trajectories. These climate differences raise the optimal social cost of carbon and underscore the need for stronger mitigation, highlighting the policy relevance of deforestation and urbanisation.

Neglecting land use dynamics therefore risks systematic underestimation of future atmospheric carbon burdens, temperature rise, and the carbon price required to meet climate targets. Carbon capture and storage experiments corroborate that higher carbon taxation (or equivalent measures) becomes necessary whenever land management is imperfect or trends towards net deforestation.

Extending the analysis with linear pattern scaling links global mean warming to regional outcomes, revealing that local temperature responses typically range from 50% to 250% of the global mean. Although this spatial resolution enriches impact assessments, it also inherits the structural uncertainty of the underlying earth system model patterns.

A distinguishing strength of the framework is its transparency. In contrast to many machine-learning emulators that replicate earth system model output, our constrained least-squares calibration enforces mass balance, non-negativity, and dynamical stability, and is accompanied by automated validation tests that expose how calibration choices propagate to economic indica-

tors. All model variants are openly available, via a simple Python API, at https://github.com/ClimateChangeEcon/Building_Interpretable_Climate_Emulators_forEconomics, enabling effort-less exploration of structural and parametric uncertainty.

Institute of Computing, Università della Svizzera italiana, Switzerland
Institute for Atmospheric and Climate Science, ETH Zürich, Switzerland
ifo Institute & Ludwig-Maximilian University of Munich, Germany
University of Zürich, Switzerland & Swiss Finance Institute (SFI), Switzerland
University of Lausanne, Switzerland, Grantham Research Institute, London School of Economics, UK & Enterprise for Society (E4S), Switzerland
Institute of Computing, Università della Svizzera italiana, Switzerland

Additional Supporting Information may be found in the online version of this article:

Online Appendix Replication Package

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